



Mahidol University
Wisdom of the Land

Pharmacology of Antithrombotic Drugs

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Department of Pharmacology
Faculty of Pharmacy
Mahidol University

Pharmacology I (PYCH304)

13th Nov 2025 (9.00-10.00)

19th Nov 2025 (8.00-10.00)



Learning Objectives

After the class, students should be able to

1. Explain the mechanism of action (MOA), pharmacokinetic and pharmacodynamic properties of antiplatelets
2. Explain ADRs and drug interactions of antiplatelets
3. Explain the MOA, pharmacokinetic and pharmacodynamic properties of anticoagulants
4. Explain ADRs and drug interactions of anticoagulants
5. Recognize the pharmacological properties of fibrinolytics



Introduction to Antithrombotics

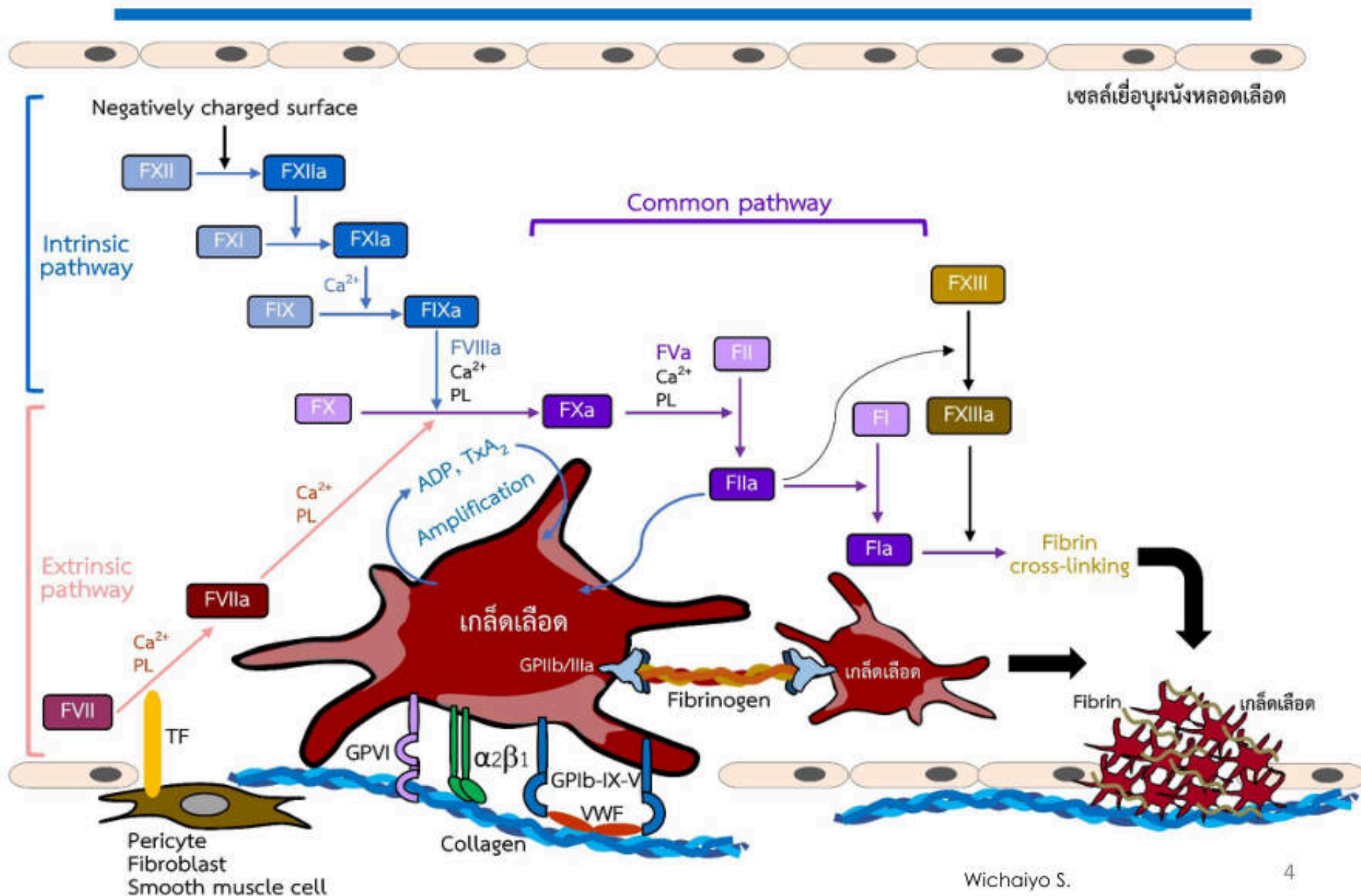
- **Thrombosis/thromboembolism**

- **Thrombus** (*thrombi*) = Blood clot that forms in a vessel and remains there
- **Embolus** (*emboli*) = A clot that travels from the site where it formed to another location in the body
- **Arterial thrombosis** is platelet-rich (called "white clot")
- **Venous thrombosis** is fibrin-rich (called "red clot")

- **Antithrombotic Drugs**

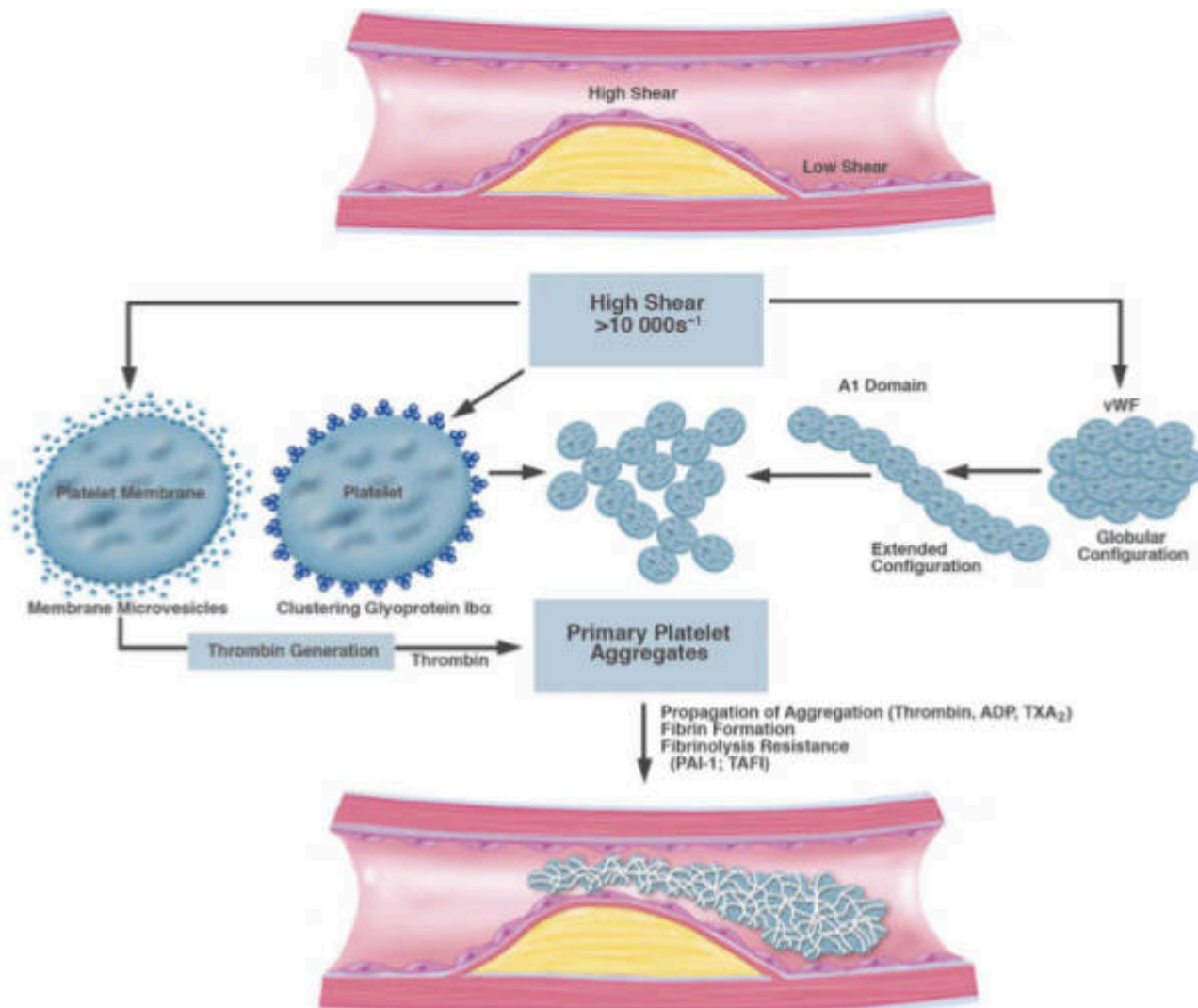
1. **Antiplatelets**
2. **Anticoagulants**
3. **Thrombolytics/Fibrinolytics**

เซลล์เยื่อบุผนังหลอดเลือด

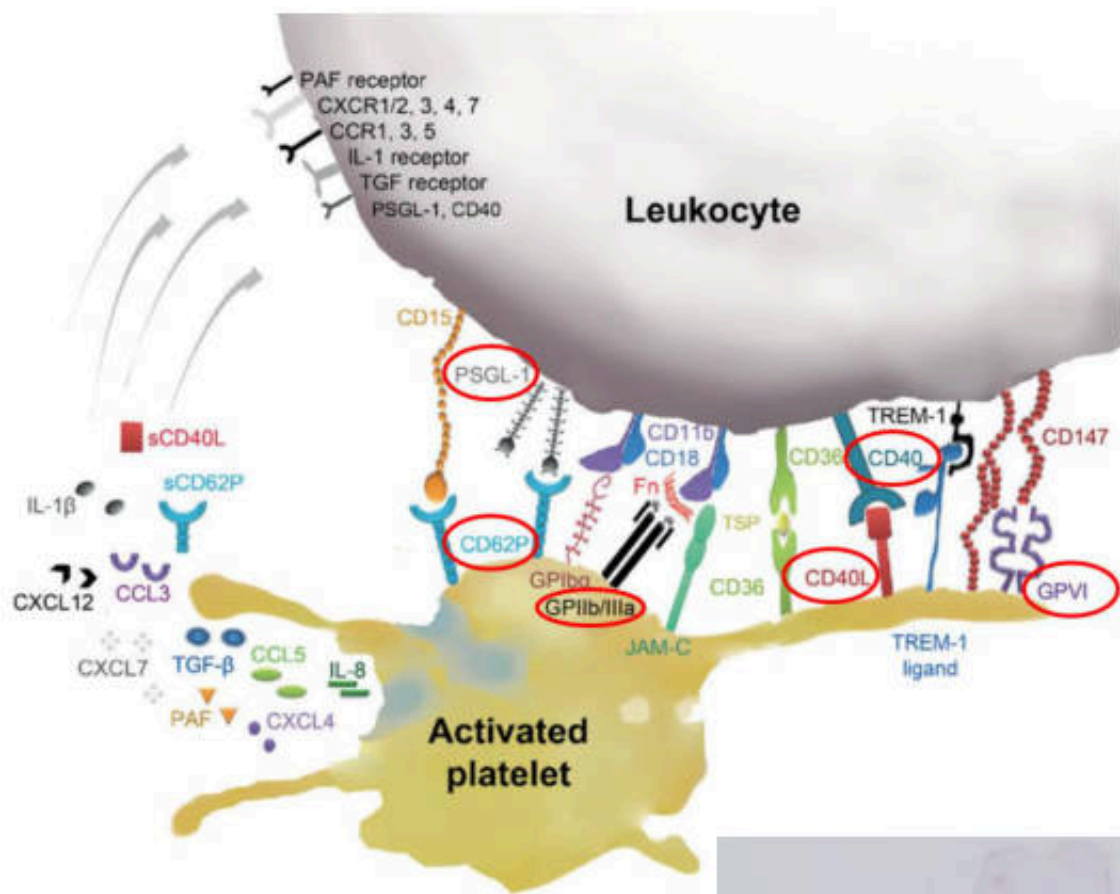




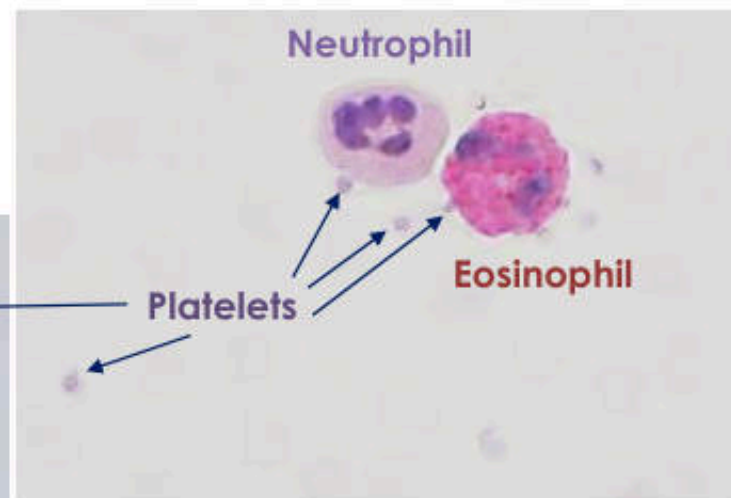
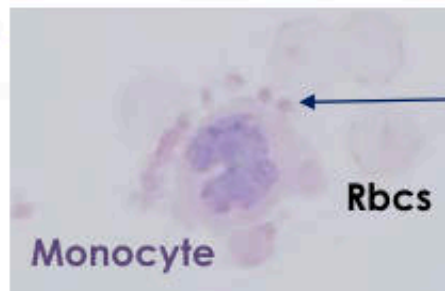
Contribution of high shear forces to arterial thrombogenesis



Role of platelets in thromboinflammation

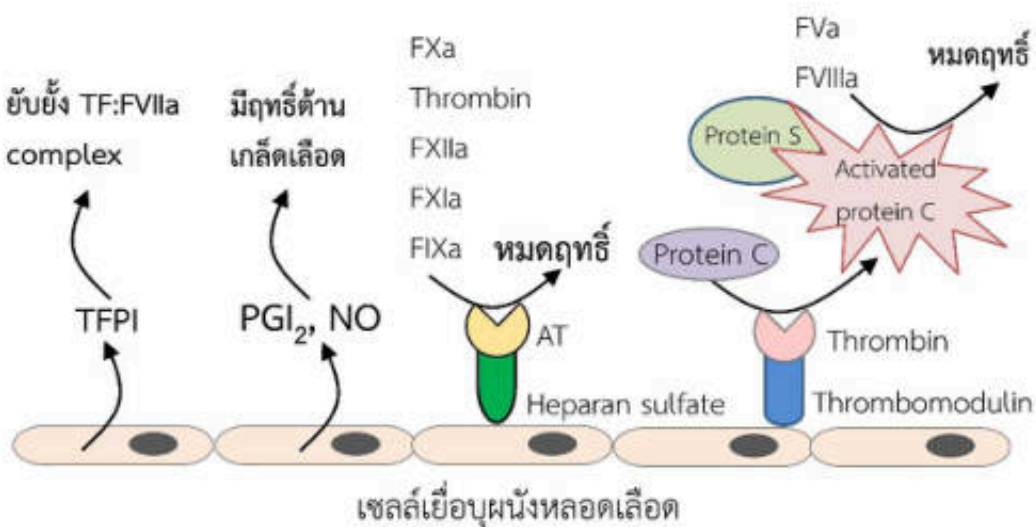


- **CD11b/CD18** = Mac-1 or $\alpha M\beta 2$ integrin
- **CD62P** = P-selectin
- **PSGL-1** = P-selectin glycoprotein ligand-1
- **CD147** = EMMPRIN (extracellular matrix metalloproteinase inducer)
- **TREM-1** = Triggering receptor expressed on myeloid cells-1



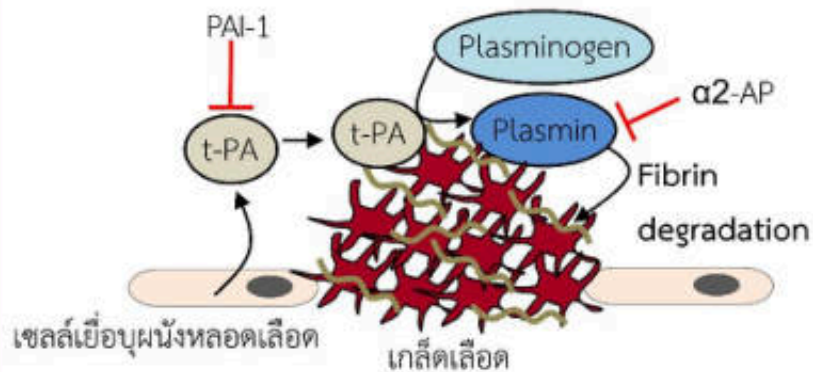
Endogenous Anticoagulation Processes

กระบวนการต้านการแข็งตัวของเลือดภายในร่างกาย



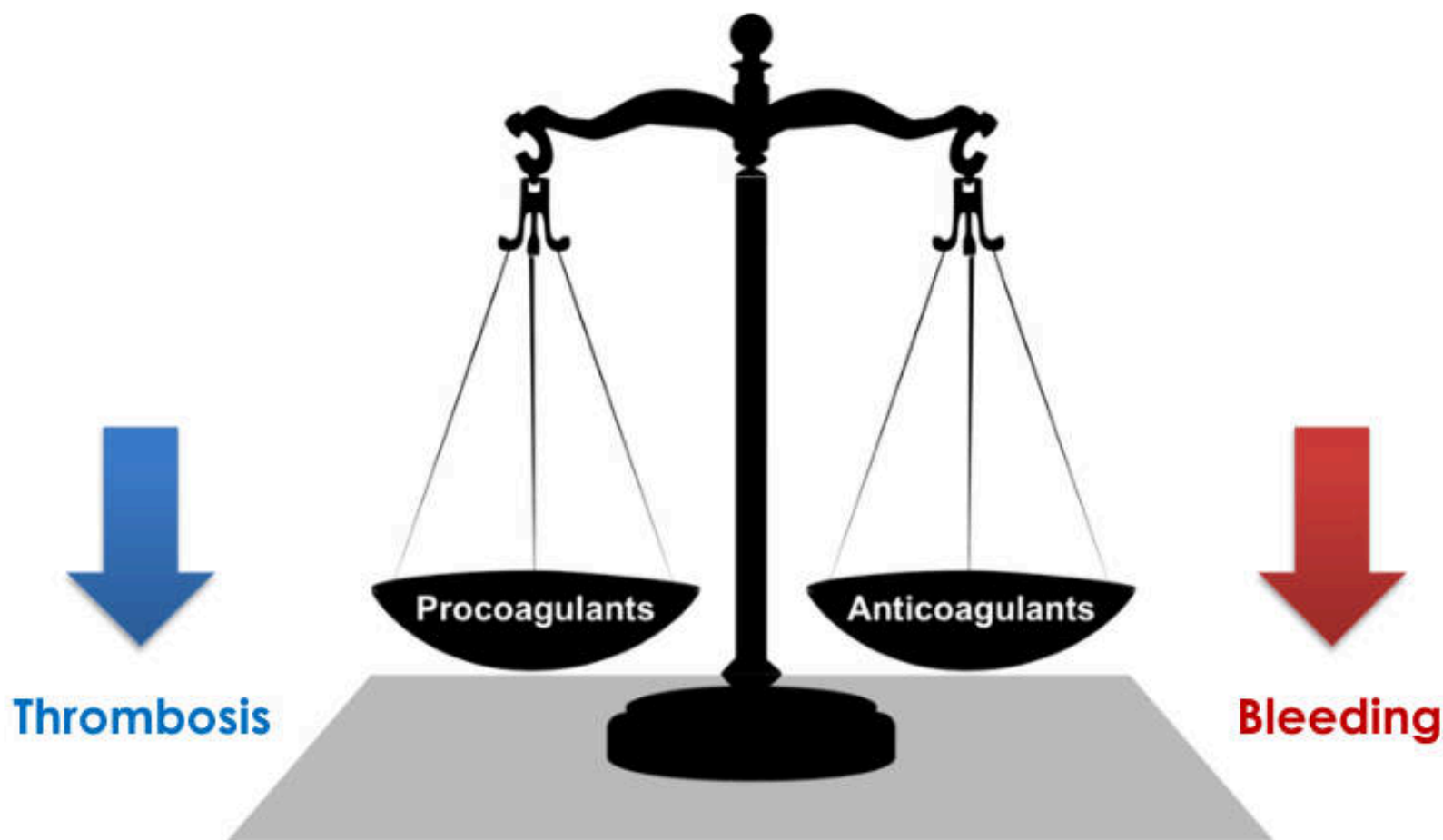
Wichaiyo S.

การสลายลิ่มเลือด





Imbalance Between Procoagulants and Anticoagulants





Skin wound in thrombocytopenic mice

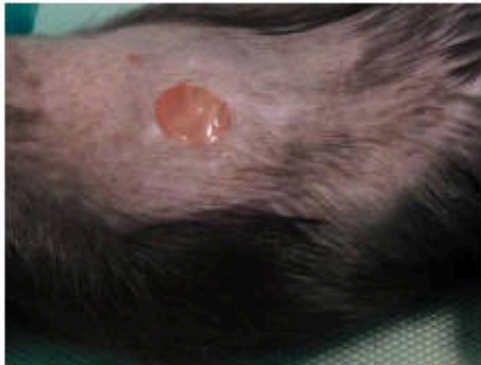
Platelet count
($\times 10^3/\text{mm}^3$)

238

68

8

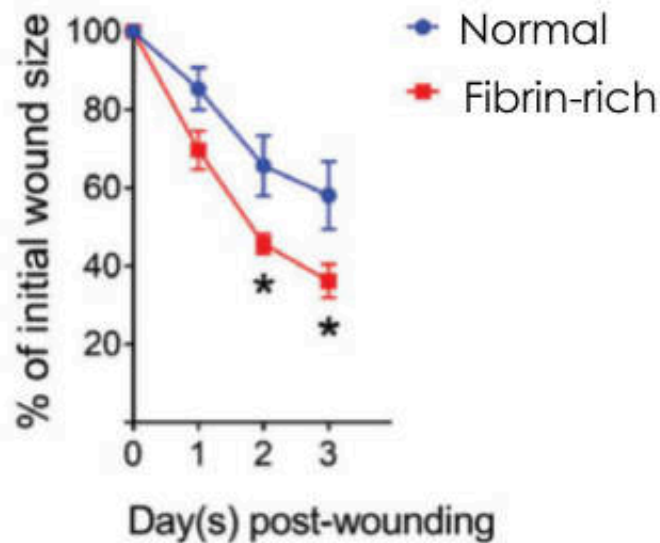
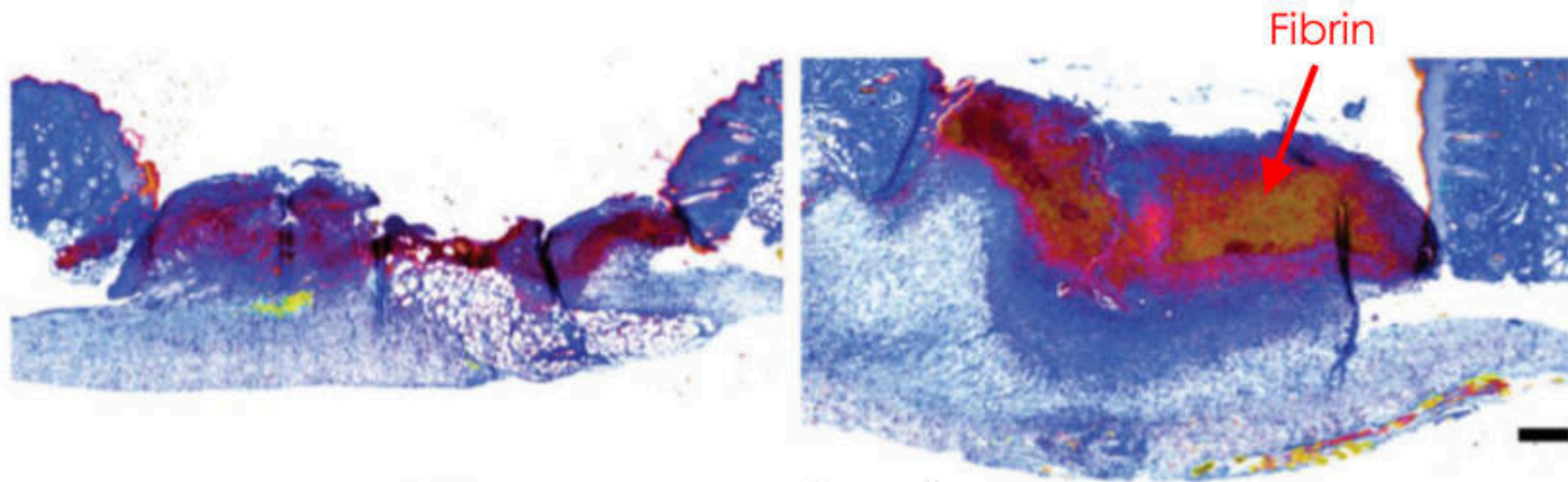
Immediately
after biopsy



2 hours
after biopsy



Association of fibrin in wound healing





Antiplatelets

- **Classified into 5 groups**
 - COX-1 inhibitors
 - ADP receptor (P2Y₁₂) antagonists
 - GPIIb/IIIa (integrin α IIb/ β III) antagonists
 - Phosphodiesterase inhibitors
 - Thrombin receptor (PAR-1) antagonists

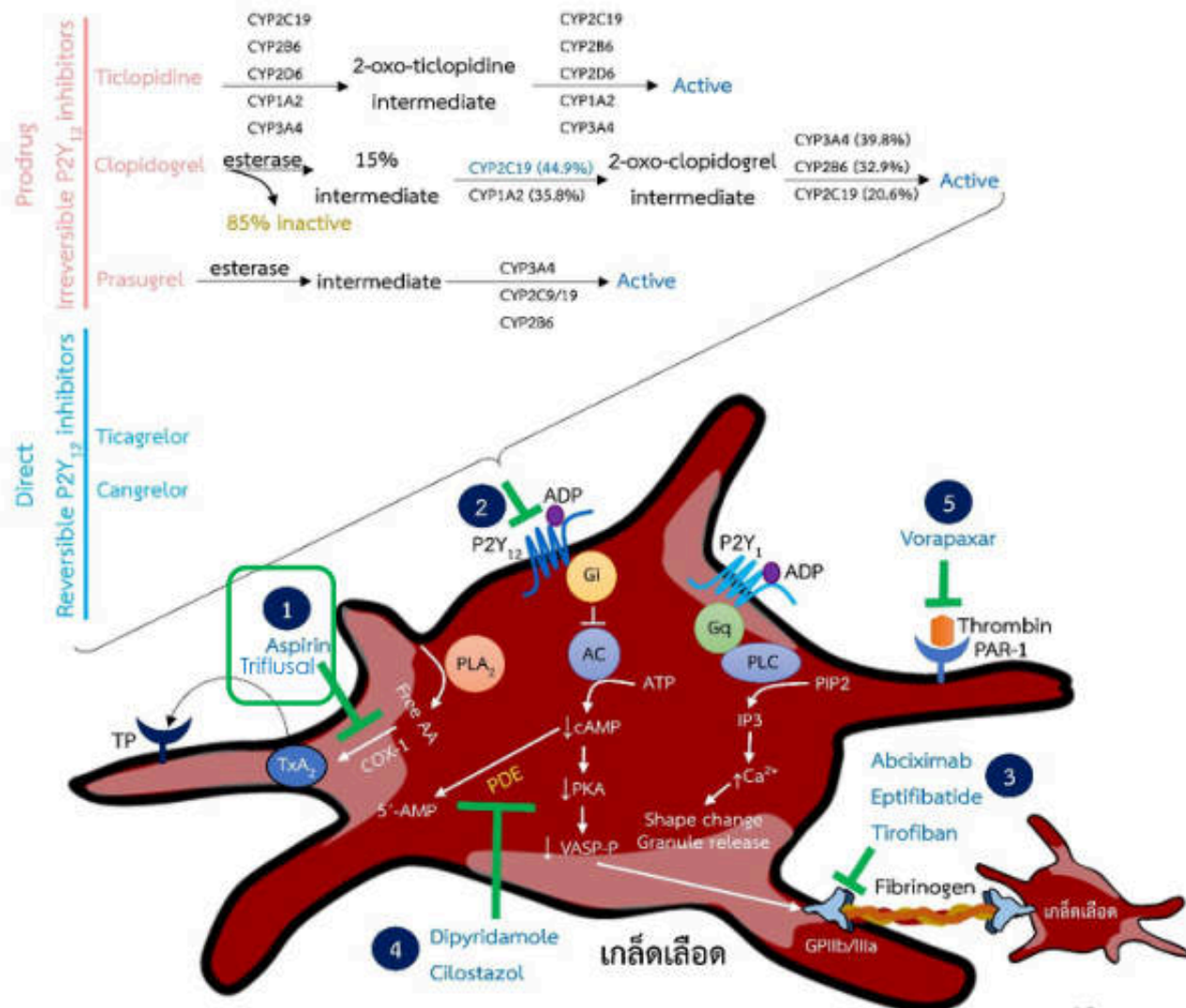
COX-1 Inhibitors

- Aspirin (ASA)**

- Low-dose
- Specifically inhibits COX-1

- Triflusal**

- COX-1 inhibition is one of the possible mechanisms





Aspirin

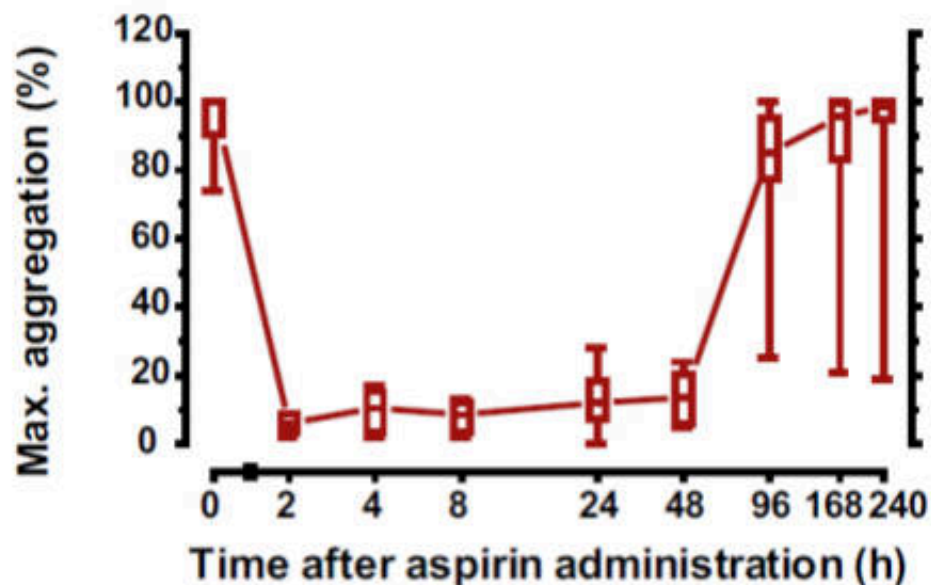
- Rapid inhibition of platelet-derived COX-1 at pre-systemic circulation
- **Short plasma $t_{1/2}$ (15-30 mins)**
- **But inhibit COX-1 throughout the lifespan of platelets (7-10 days)**
 - Selective **irreversible inhibition**
- **However, 10-15% of platelets is synthesized daily**
 - After 48 h of aspirin discontinuation, 20-30% of platelets respond to TxA_2 normally
 - **Support a daily dosing, and no need for discontinuation before most of elective surgeries**



Pharmacodynamic effect of aspirin

- Aspirin inhibits arachidonic acid-induced platelet aggregation by ~70-90%

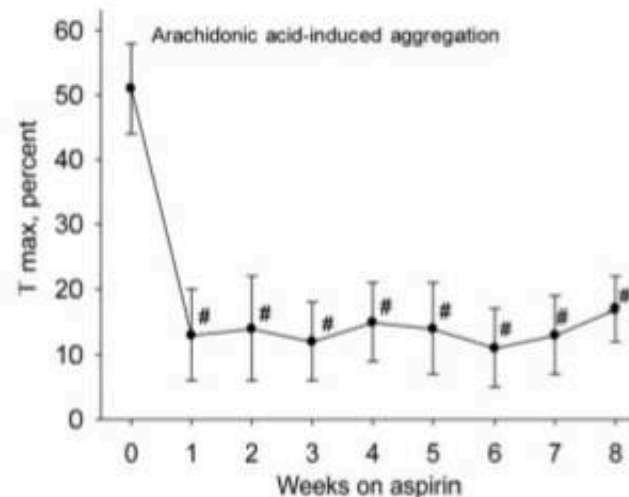
a single dose of 325 mg aspirin



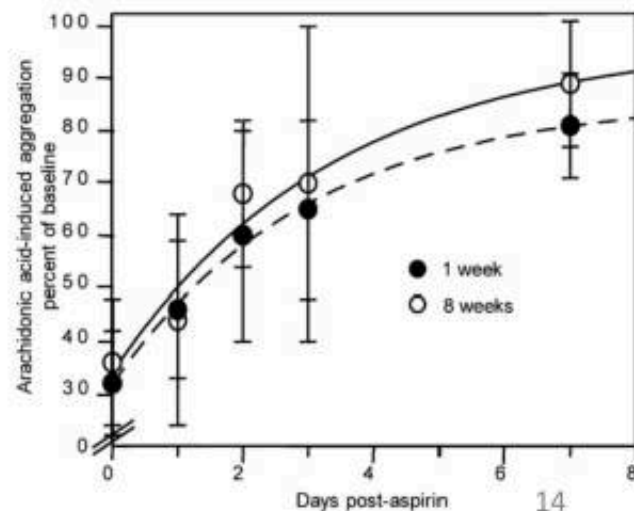
Proc Natl Acad Sci U S A 2014;111(47):16830-5.

Low-dose aspirin

Maximal aggregation (Tmax) values



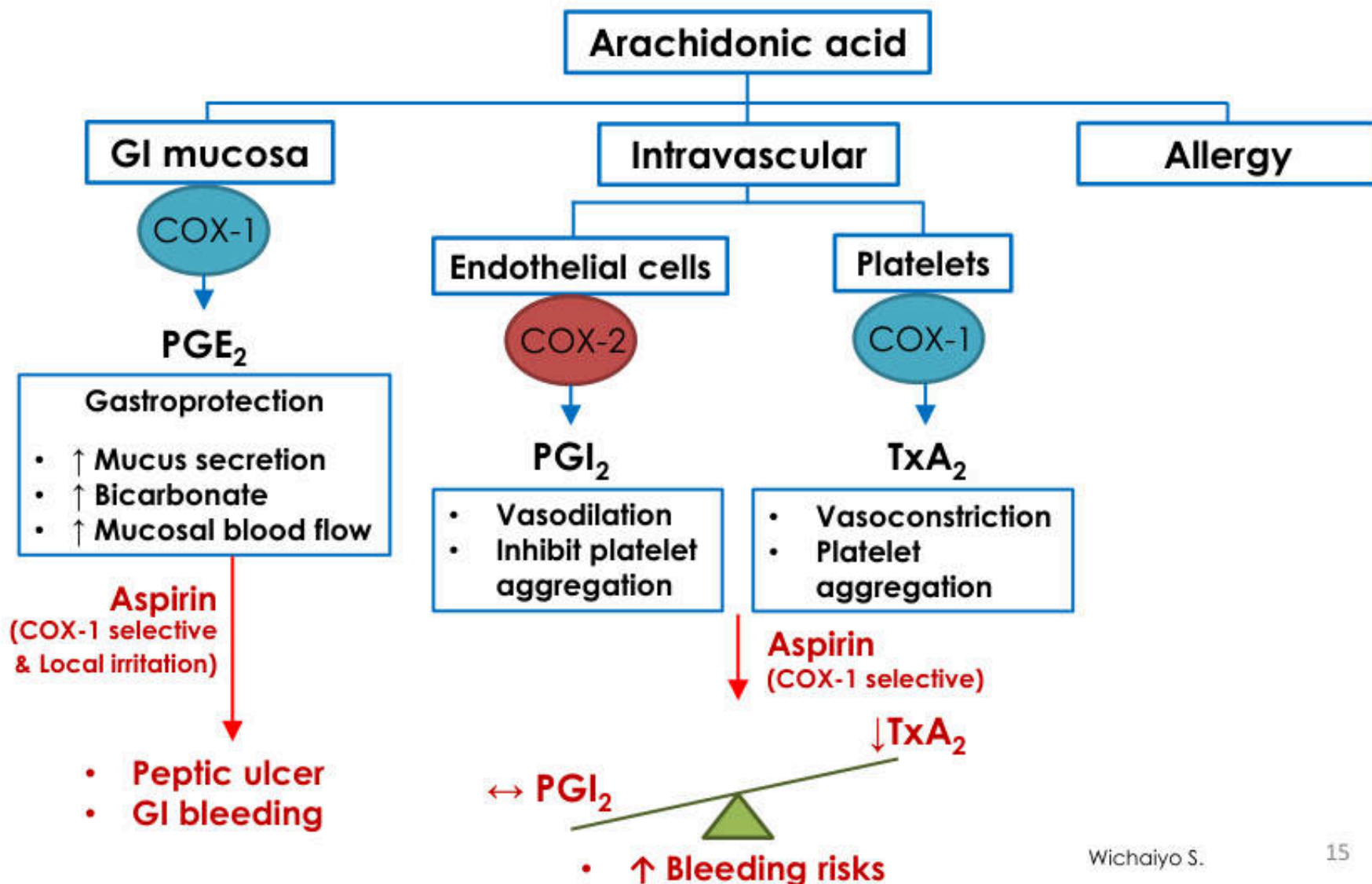
Recovery of platelet aggregation after aspirin



J Am Coll Cardiol. 2009;53:667-77.



Adverse Effects of Aspirin



P2Y₁₂ Antagonists

• Thienopyridines

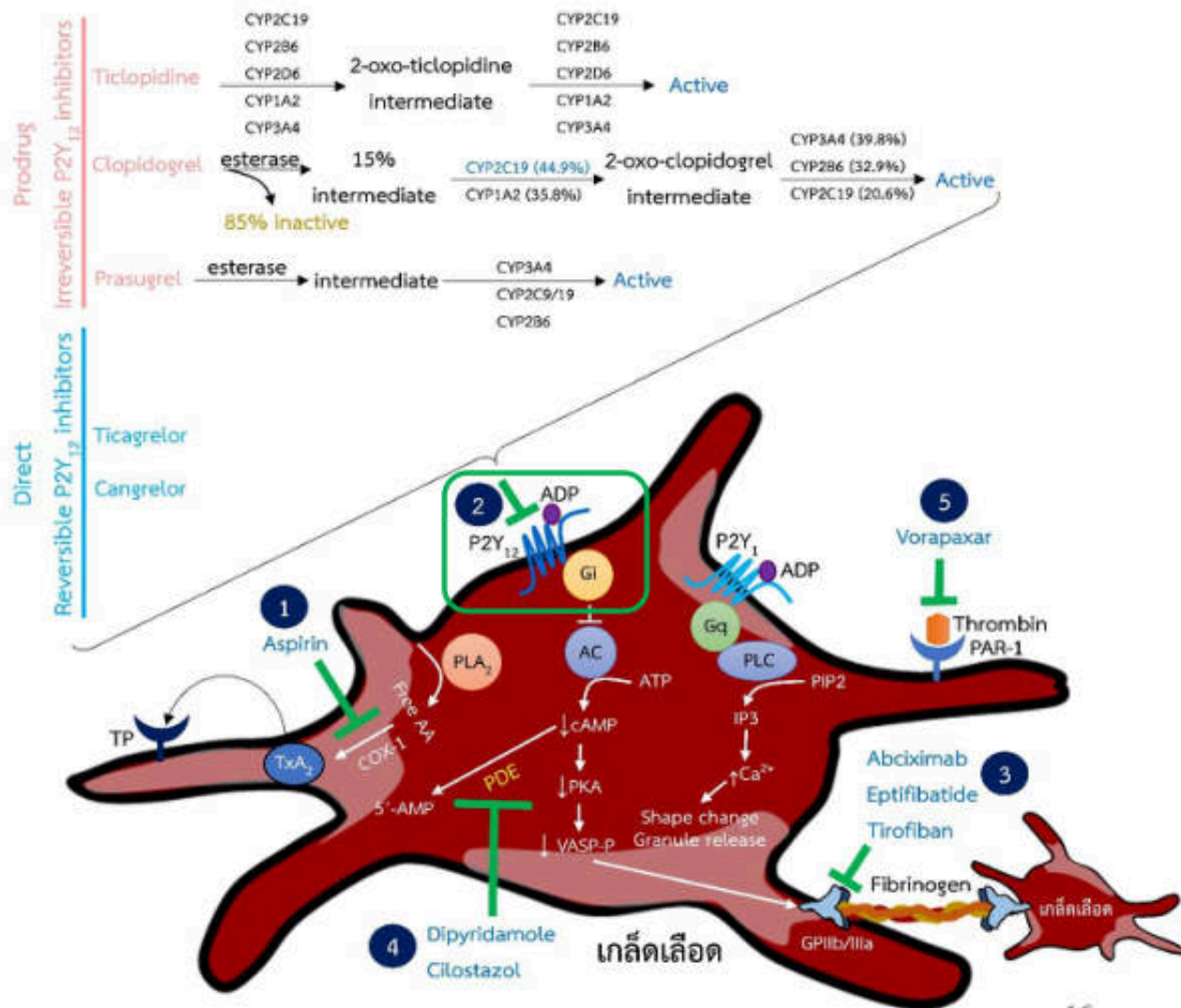
- Irreversible P2Y₁₂ inhibition

- Ticlopidine
- Clopidogrel
- Prasugrel

• Non-thienopyridines

- Reversible P2Y₁₂ inhibition

- Ticagrelor
- Cangrelor





Ticlopidine

- The first Thienopyridines
- **Inhibits ADP-induced platelet aggregation by 40-50%**
 - Normal dosing (250 mg BID) takes 3-7 days to reach this effect
- Non-linear PK
 - Initial half-life: 12.6 h
 - Long-term use leads to longer half-life (4-5 days)
- Common ADRs: Nausea, Vomiting, Diarrhea
- **Serious ADRs (resolved with drug discontinuation)**
 - Neutropenia
 - Aplastic anemia
 - Thrombotic thrombocytopenic purpura (TTP)
 - Elevated liver enzymes



Clopidogrel

- Inhibits ADP-induced platelet aggregation by 40-50% (same as ticlopidine)
 - Loading dose (300-400 mg) takes 2 h
 - Normal dose (75 mg OD) takes 3-7 days
- CYP2C19 polymorphisms with reduced enzyme activity can attenuate drug responses
 - CYP2C19*2 (681G → A)
 - CYP2C19*3 (636G → A)
- CYP2C19 inhibitors e.g. omeprazole and esomeprazole reduce antiplatelet activity
- ADRs
 - Lower bleeding risk than aspirin
 - Lower blood dyscrasia than ticlopidine

CYP2C19 polymorphisms in the Thai population and the clinical response to clopidogrel in patients with atherothrombotic-risk factors

Chonlaphat Sukasem¹
 Ramaimon Tunthong²
 Montri Chamnanphon¹
 Siwalee Santon¹
 Thawinee
 Jantararoungtong¹
 Napatrupron Koomdee¹
 Santirhat Prommas¹
 Apichaya Puangpetch¹
 Prin Vathesatogkit²

Predicted phenotype	Genotypes	Samples (number)	Frequency (%)
Extensive metabolizer	*1/*1	428	40.72
Intermediate metabolizer	*1/*2	369	35.10
	*1/*3	72	6.85
Poor metabolizer	*2/*2	77	7.32
	*2/*3	59	5.61
	*3/*3	1	0.10
Ultra-rapid metabolizer	*1/*17	45	4.30
Total		1,051	100.00

Abbreviation: *CYP2C19*, cytochrome P450 2C19.

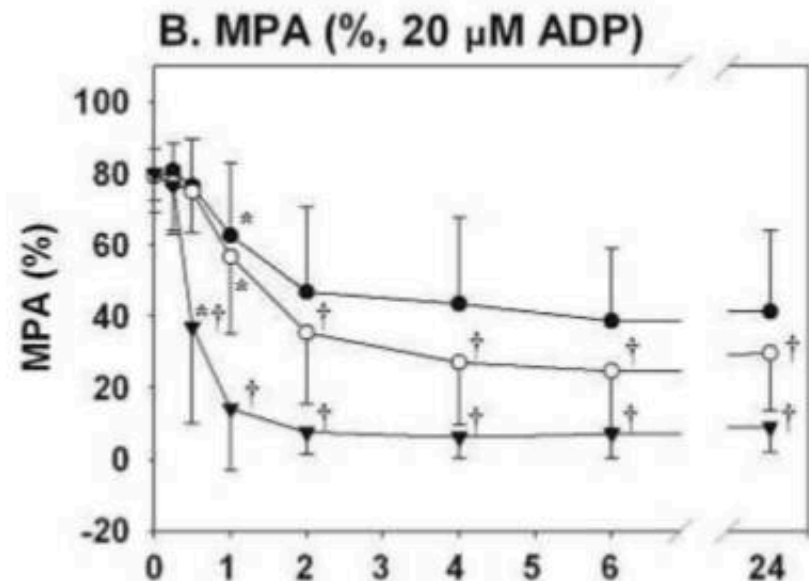
Predicted phenotypes	Responders n = 71 (%)	Non-responders n = 25 (%)	P-value*
Extensive metabolizer	32 (45.1)	9 (36.0)	0.304
Intermediate metabolizer	26 (36.6)	7 (28.0)	0.435
Poor metabolizer	11 (15.5)	9 (36.0)	0.030
Ultra-rapid metabolizer	2 (2.8)	0 (0.00)	–

Note: *P-values were determined by a chi-square test.



Prasugrel

- **More potent and rapid than clopidogrel**
 - Inhibit ADP-induced platelet aggregation by **70-80%**
 - Loading dose (60 mg) takes 2 h
 - Significantly different effect is observed at 30 min
- ADRs
 - **Higher bleeding risk than clopidogrel**
 - **Contraindicated in patients with prior stroke or TIA or ≥ 75 yrs**
 - Lower TTP than ticlopidine



MPA = Maximal platelet aggregation

clopidogrel 300 mg (●), clopidogrel 600 mg (○), and prasugrel 60 mg (▼) doses

Ticagrelor

- More potent and rapid ON/OFF effect relative to clopidogrel**

- Inhibit ADP-induced platelet aggregation by **80-90%**

- Loading dose (180 mg) takes 2 h
 - Significantly different effect is observed at 30 min

- Non-competitive reversible P2Y₁₂ inhibitor**

- Bind at allosteric site or act as inverse agonist

- Also inhibit reuptake of adenosine**

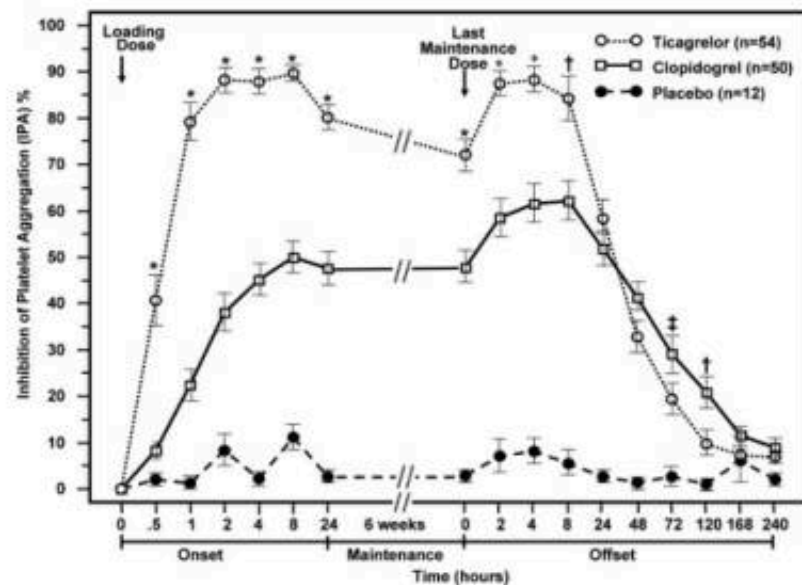
- Metabolized by CYP3A4 & excreted via feces

- Half-life: 6-8 h (active met. 8-12 h)

- Unique ADR: Dysnea, dizziness, headache**

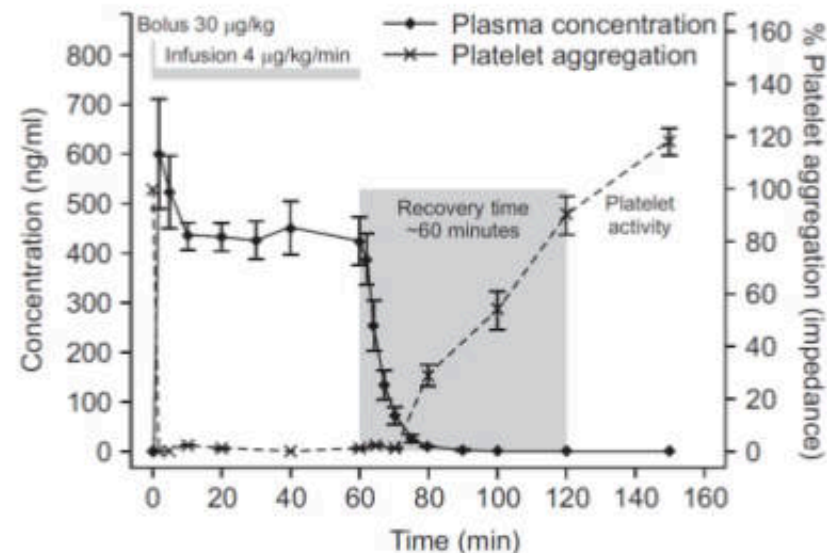
- Possibly related to adenosine

- Bleeding risk is comparable with clopidogrel**



Cangrelor

- ATP analogue
- The **only injectable drug** in this group
- Competitive reversible P2Y₁₂ inhibitor
- **More potent and rapid ON/OFF effect compared with clopidogrel**
 - **Inhibit ADP-induced platelet aggregation by ~100%**
 - Loading dose (30 µg/kg) takes < 5 min
- Metabolized by dephosphorylation
- Very short half-life: 3-9 min
- **Duration: 1-2 h (Benefit for bridging therapy)**
- Comparable bleeding risk with clopidogrel



Korean J Anesthesiol 2017;70(4):379-389.

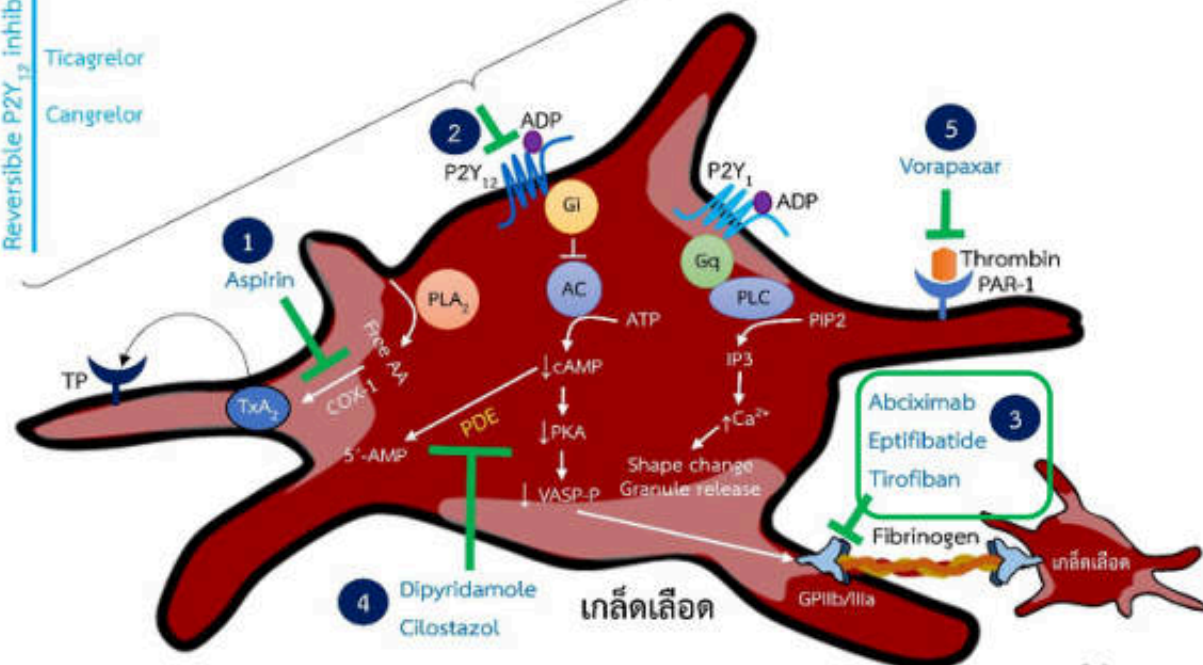
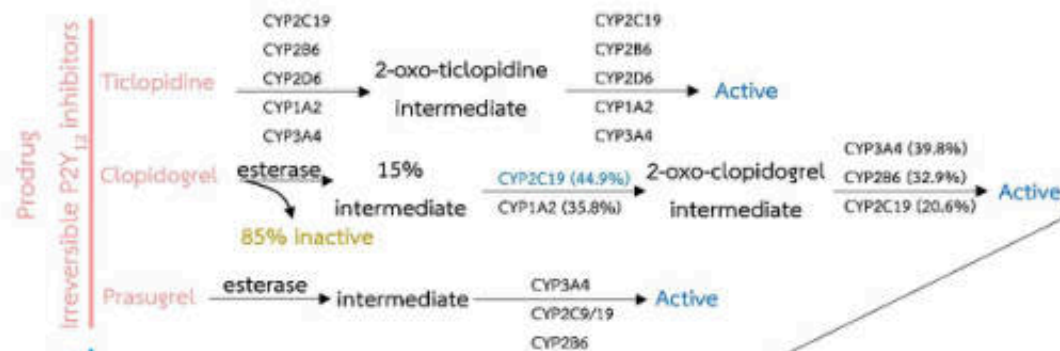


Comparative Pharmacological Properties of P2Y₁₂ antagonists

	Clopidogrel	Prasugrel	Ticagrelor	Cangrelor
Maximal inhibition of platelet aggregation (%)	40-50	70-80	80-90	100
Bioavailability (%)	50	80	36	100
Half-life (active met.)	30-60 min	Distribution half-life 30-60 min; elimination half-life: 2-15 h	7-9 h (active met. 12 h)	3-6 min
Binding reversibility	Irreversible	Irreversible	Reversible	Reversible
Onset of action (loading)	2-6 h	30 min	30 min	2 min
Frequency of administration	OD	OD	BID	i.v. infusion
Duration of effect	3-10 days	7-10 days	3-5 days	1-2 h
Clinical indication	PCI, ACS, ACS-PCI, and SCAD	ACS-PCI	ACS, ACS-PCI	PCI, bridging in patient at high ischemic risk who is undergoing an invasive procedure
Discontinuation before non-acute surgery	At least 5 days	At least 7 days	At least 3 days	1 h

GPIIb/IIIa Antagonists

- GPIIb/IIIa antagonists have no effect on platelet signaling/activation





Comparative Pharmacological Properties of GPIIb/IIIa antagonists

Parameter	Abciximab	Eptifibatide	Tirofiban
Molecular weight (D)	47615	800	495
Receptor selectivity			
GP IIb/IIIa ($\alpha_{IIb}\beta_3$)	+++	+++	+++
$\alpha_v\beta_3$	+++	+	0
$\alpha_M\beta_2$ (MAC-1)	++	0	0
Physical binding site at GPIIb/IIIa	RGD-sequence binding pocket + secondary fibrinogen Binding site (KQAGDV)	KGD-sequence	RGD-sequence
Stoichiometry of binding	1.5–2	>250	250–2500
k_D (nM)	5	15	120
Clearance	Platelet binding, unbound substance via proteolytic cleavage	Renal (60–70%) Biliar (20–30%)	Renal (98%) Partially metabolized
Volume of distr. (l/kg)	ca. 0.07	0.2–0.3	0.2–0.3
Half-life			
Plasma	ca. 20–30 min	ca. 1.5–2.0 h	ca. 2–3 h
Platelets	ca. 4 h	10–15 s	10–15 s
Restoration of normal Hemostatic function	ca. 72 h	3–4 h	4 h

+++ = strong activity, ++ = moderate activity, + = weak activity, 0 = no activity. ca. = circa (about), k_D = dissociation constant.



GPIIb/IIIa Antagonists (cont.)

- **Abciximab**
 - Fab fragment of **humanized chimeric** monoclonal Ab
 - **Hypersensitivity reaction**
 - **Act as steric hindrance** (not competitively bind to GPIIb/IIIa)
- **Eptifibatide** (cyclic heptapeptide)
 - Competitive binding to GPIIb/IIIa
 - Excreted via kidney as unchanged (98%)
- **Tirofiban** (tyrosine derivative)
 - Competitive binding to GPIIb/IIIa (same as eptifibatide)
 - 60-70% excreted via kidney as unchanged

PDE inhibitors

• PDE isozymes

– PDE-3

degrades cAMP (e.g. in platelets)

– PDE-5

degrades cGMP (e.g. in vascular smooth muscle cells)

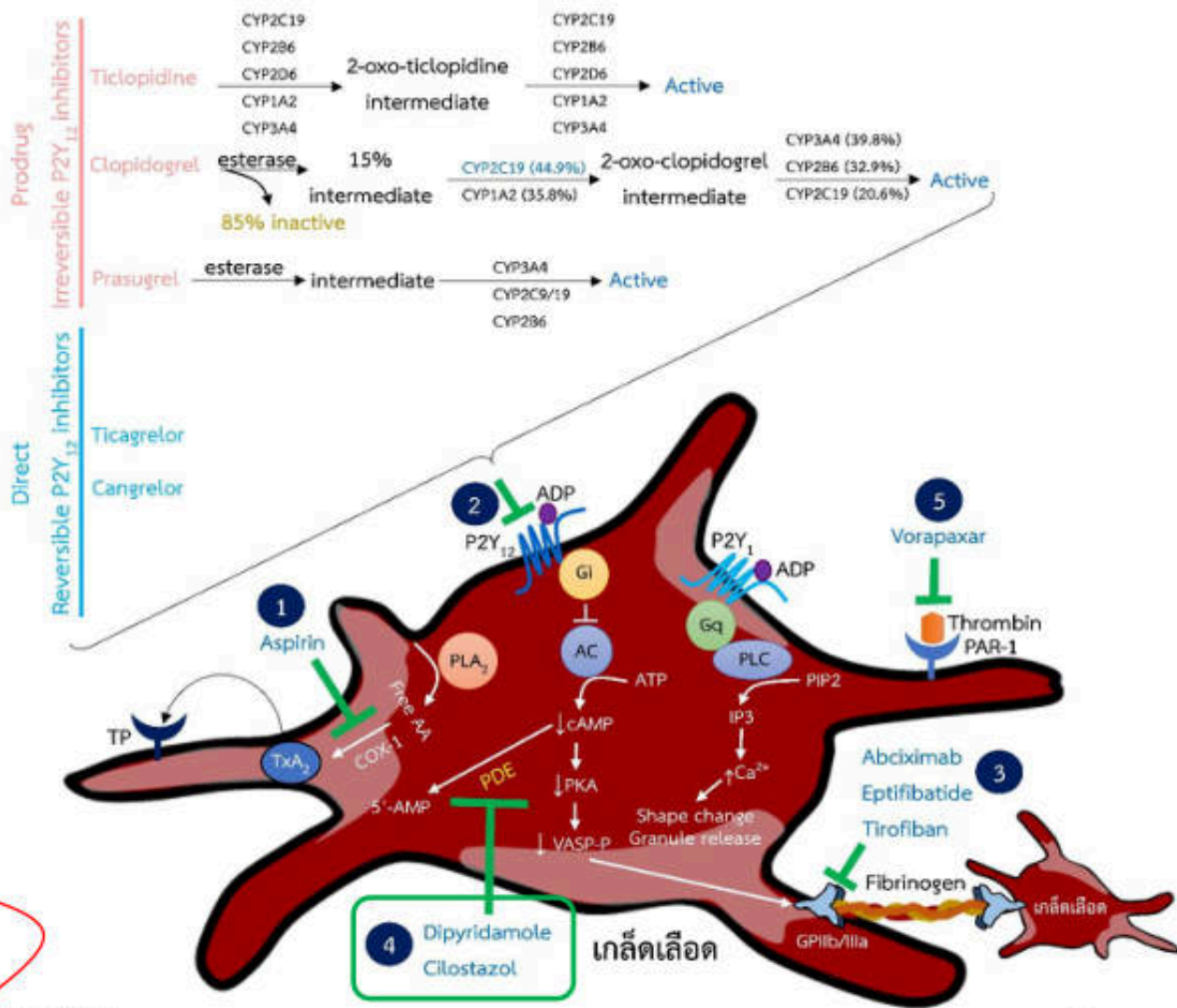
• PDE inhibitors

– Dipyridamole

– Cilostazol

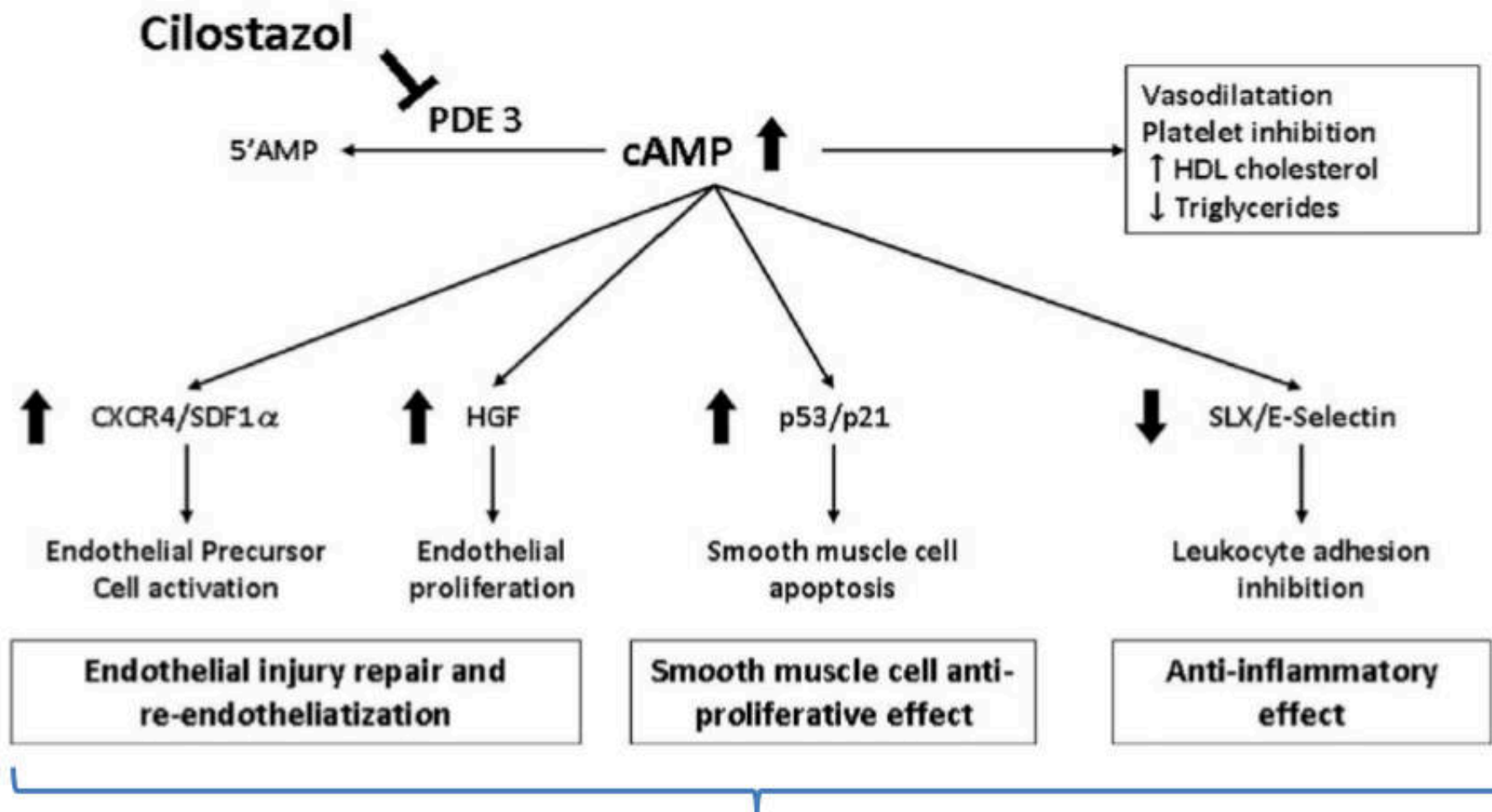
Used in intermittent claudication

Have common role in 2nd prevention of stroke





Other potential activities of cilostazol



Possibly support the role of cilostazol in patients with intracranial arterial stenosis



PDE inhibitors (cont.)

- **Dipyridamole**

- 
- Inhibit both PDE-3 and PDE-5
 - Also inhibit reuptake of adenosine
 - Antiplatelet and vasodilatory effects
 - Metabolized by glucuronidation (Half-life: 10 h)
 - Common ADR: Headache (at beginning)

- **Cilostazol**

- 
- Potent PDE-3 inhibitor
 - Also inhibit reuptake of adenosine
 - Antiplatelet and vasodilatory effects
 - Metabolized by CYP3A5 (Half-life: 10 h)
 - ADRs: Headache, loose stool, palpitation
 - **Also used to treat intermittent claudication in PAD**



Tissue distribution of PDE3 and PDE5



PDEs Family	Substrate	Primary Tissue Distribution of PDEs	Biological Effects of PDEs	Examples of Inhibitors
PDE3	cAMP/cGMP	<u>heart</u> , <u>smooth muscle</u> , lung, liver, platelets, <u>adipocyte</u> , pancreas, immune cells, brain, endothelium, epithelium, oocytes	<u>myocardial contractility</u> , platelet aggregation, vascular and airway <u>smooth muscle contraction</u> , regulation of inflammation and fibrosis, platelet functions, mediation of response to insulin, <u>regulation of cell proliferation</u>	milrinone cilostazol olprinone ensifentrine (RPL554)
PDE5	cGMP	lung, platelets, smooth muscle, heart, endothelial cells, brain, kidney, gastrointestinal system	vascular and airway smooth muscle contraction and remodeling, platelet functions, participation in inflammation and oxidative stress, neuronal functions	sildenafil dipyridamole zaprinast tadalafil vardenafil

Int. J. Mol. Sci. 2021, 22, 1929. <https://doi.org/10.3390/ijms22041929>

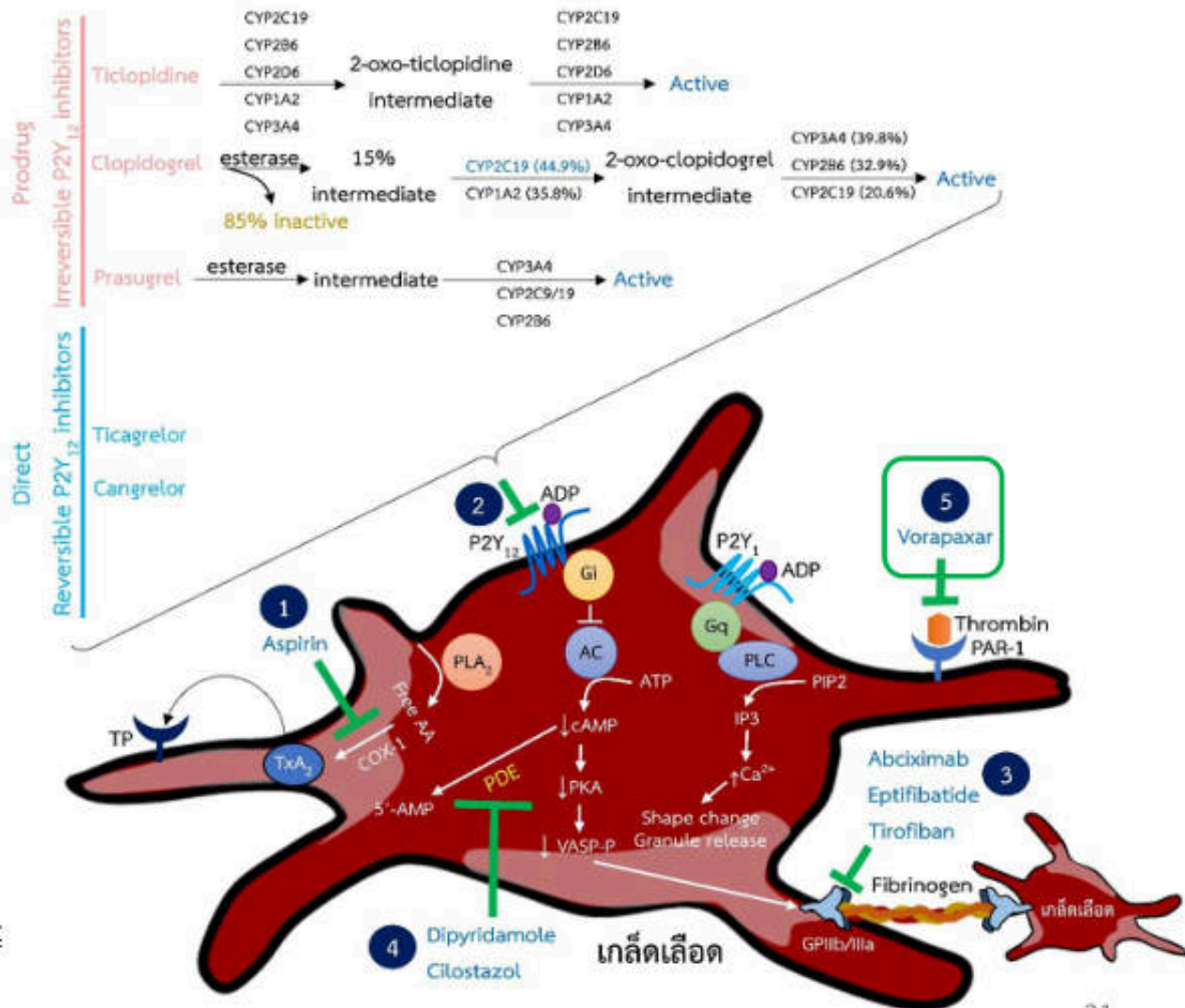
PAR-1 Antagonists

Receptor	Cells	Physiological role
PAR-1	Platelets ⁺ Endothelium SMCs Leukocytes	Platelet activation and thrombosis Embryogenic development Vasoregulation Tissue remodelling
PAR-2	Endothelium SMCs Leukocytes	Inflammation Vasoregulation
PAR-3	Platelets [#] Endothelium	Platelet activation and thrombosis in mice
PAR-4	Platelets Endothelium SMCs Leukocytes	Platelet activation and thrombosis Inflammation

⁺human platelets
[#]mice platelets

Thromb Haemost 2008;99: 305–315.

- **PAR-1 mediates platelet activation at low thrombin concentrations**
- **PAR-4**, that lacks the thrombin binding site, is only able to **mediate platelet activation at high thrombin concentrations**





Vorapaxar

- **Reversible PAR-1 antagonist**
 - **Slow dissociation from receptor**
 - Inhibit thrombin receptor-activating peptide-induced platelet aggregation by 80% at 2 h
- Does not affect enzymatic activity of thrombin
- Metabolized by CYP3A4
- **Half-life: 150-300 h**
- **FDA approved (but not as 1st line) for 2nd prevention in patients with prior MI and PAD**
- **Contraindicated in patients with prior stroke**



Anticoagulants

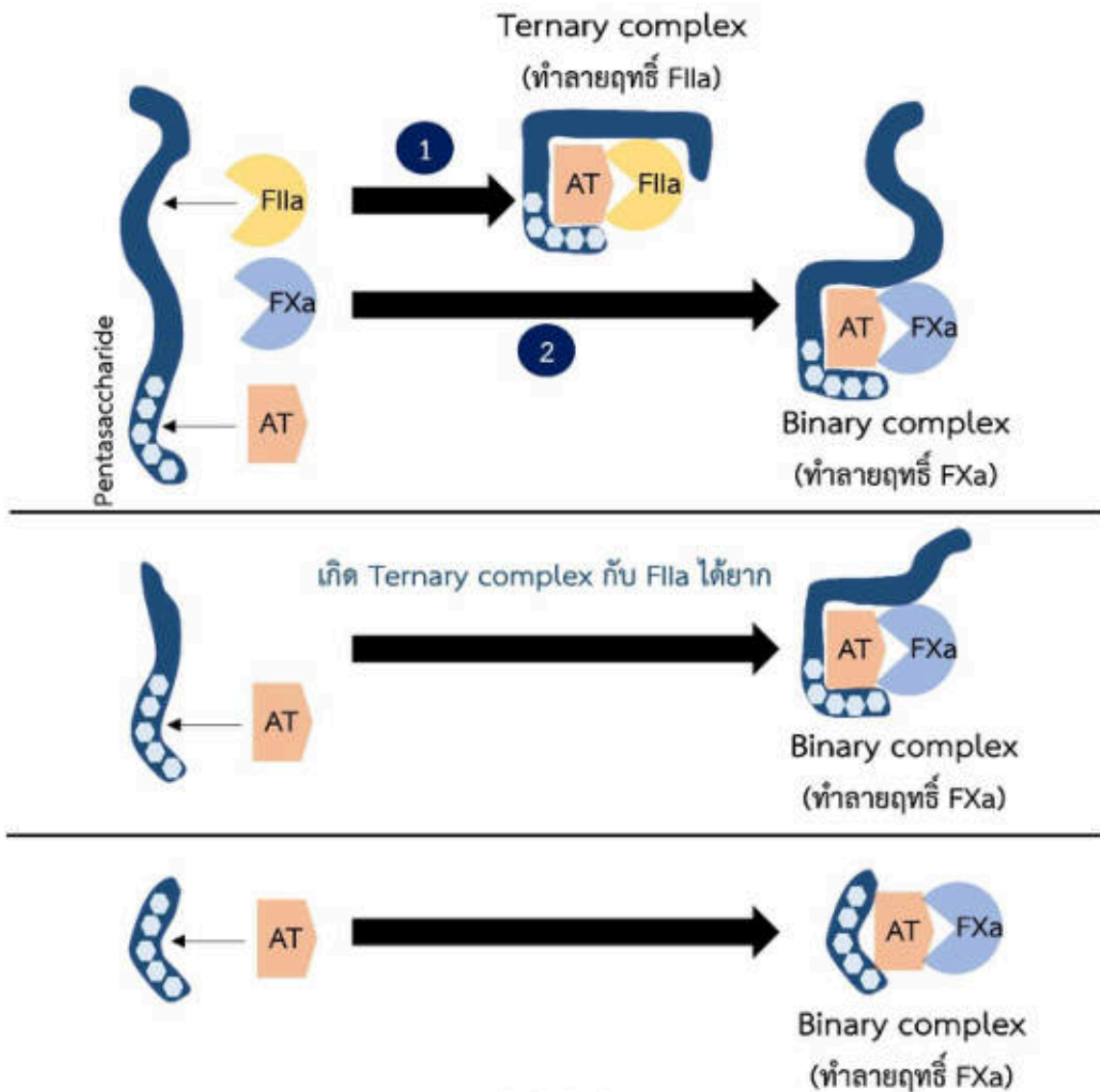
Indirect-acting anticoagulants

1. UFH
2. LMWH
3. Fondaparinux
4. Warfarin

Direct-acting anticoagulants

1. Thrombin inhibitors
2. FXa inhibitors

Heparin and Its Derivatives



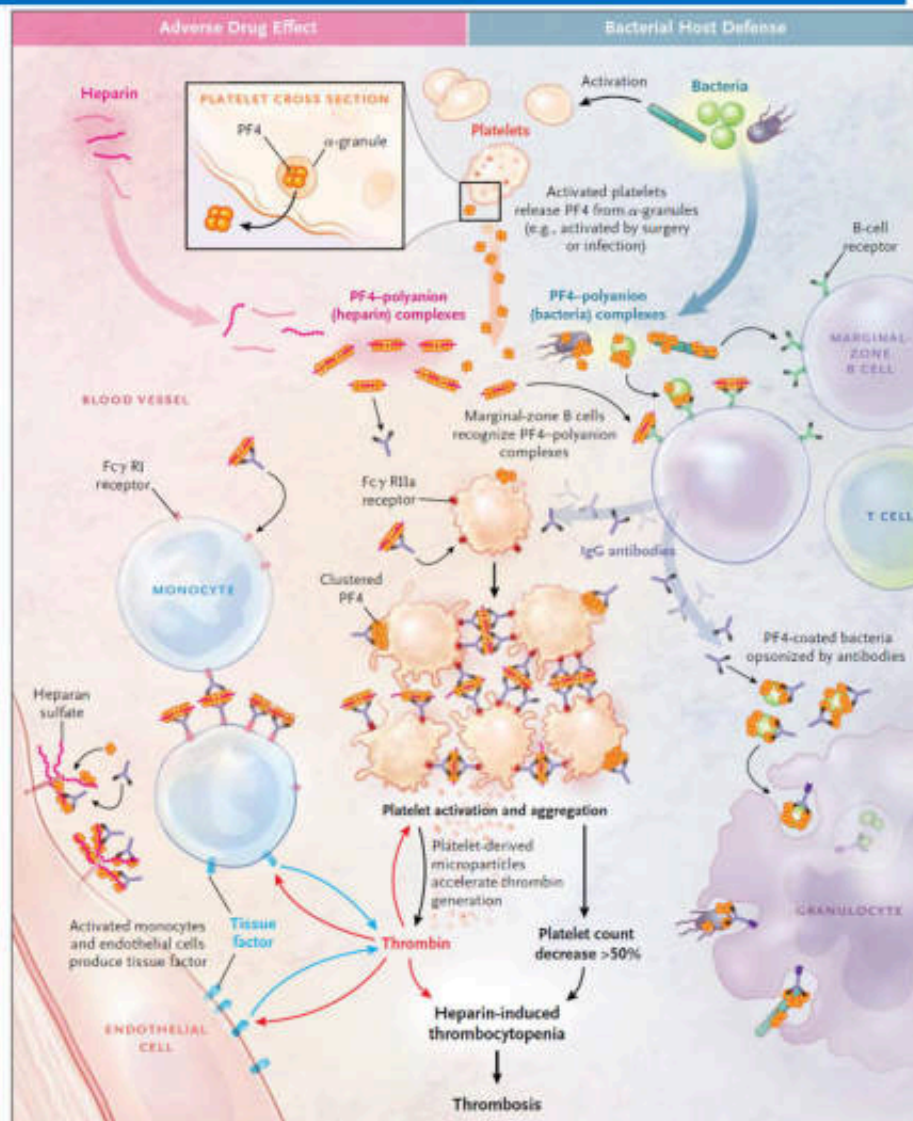


Unfractionated Heparin (UFH)

- Sulfated polysaccharides 45 units (3,000-30,000 kDa)
 - 1/3 of its structure has **antithrombin (AT) binding site**
- Inhibits FXIIa, FXIa, **FXa**, FIXa, and thrombin (**FIIa**)
- Pharmacokinetics
 - Highly bound to plasma protein
 - **Half-life: 30 mins-1.5 h (at high dose ~2.5 h)**
 - Metabolisms
 - Low-dose: quickly degraded by plasma enzymes and cells (e.g. endothelial cells and macrophages)
 - Saturable zero-order kinetics
 - **High-dose: degraded by kidney (non-saturable)**

Unfractionated Heparin (UFH)

- **Monitor aPTT (target: 1.5-2.5 x control)**
 - Intrinsic pathway
- ADRs
 - Bleeding
 - **HIT**
 - Osteoporosis
 - Long-term > 1 mo.





Low Molecular-Weight Heparin (LMWH)

- **More predictable action/effect than UFH**
- **Do not need laboratory test to monitor their effect**
- Good bioavailability (70-90%) after s.c. injection
- Short plasma half-life (3-4 h)
 - **But, mean residence time for anti-FXa: 5-7 h**
 - **Also stimulates secretion of TFPI from endothelial cells**
 - BID or OD dosing
- Enoxaparin i.v. loading is used in STEMI
- Lower rate of HIT (but can cross-react with UFH)



Fondaparinux

- Synthetic short pentasaccharide
 - AT binding site only
- Specifically inhibits FXa
- **Bioavailability 100% after s.c. injection**
- **Half-life: 17-21 h**
- Offset of action: 2-4 days after discontinuation
- HIT is very rare

Comparative Pharmacological Properties of Heparin and Its Derivatives

	UFH	LMWH			Fondaparinux
		Enoxaparin	Nadroparin	Dalteparin	
Target	FXIIa, FXIa, FXa, FIXa, FIIa	FXa > FIIa	FXa > FIIa	FXa > FIIa	FXa
Anti FXa:FIIa ratio	1.0	3.9	3.5	2.2	-
Bioavailability (%)	30	91	67	83	100
Half-life (h)	1	4.5	3	2.3	17
Renal excretion as active form	No	Yes	Yes	Yes	Yes
Antidote	Protamine	Protamine (60% effect)	Protamine (60% effect)	Protamine (60% effect)	-
HIT (%)	< 5	< 1	< 1	< 1	Case report



Warfarin

เข้าสู่กระแสเลือด

Pro-coagulants:

FII, FVII, FIX, FX

Anticoagulants:

Protein C, S, Z

γ -glutamyl
carboxylase

Non-functional
pro-zymogens

Vitamin K epoxide
(Oxidized vitamin K)

Vitamin K hydroquinone
(Reduced vitamin K)

Vitamin K-epoxide
reductase (VKORC1)

Vitamin K reductase
(Quinone reductase)

Vitamin K quinone
(Vitamin K1)



Warfarin

Wichaiyo S.

Factor	Plasma half-life (h)
FII	42-72
FVII	4-6
FIX	21-30
FX	27-48
Protein C	8
Protein S	60
Protein Z	40-45



Warfarin (Cont.)

- **Onset of action: 3-7 days**
 - Rapid-acting anticoagulants are needed at initial period
- **Monitor INR (international normalized ratio)**
 - Extrinsic pathway
 - Target range: 2-3
- **Antidote: Vitamin K 1 (Phytonadione)**
- **Other drugs used for warfarin-induced severe bleeding**
 - **3-PCCs, 4-PCCs**, FFP, rFVIIa (Novoseven)

Pharmacokinetic and pharmacodynamic properties of warfarin enantiomers

	R-warfarin	S-warfarin
Bioavailability	95–100 %	95–100 %
Volume of distribution	0.12–0.22 L/kg	0.11–0.19 L/kg
Protein binding	98.7–99.9 %	98.9–100 %
Elimination half-life	45 h (20–70 h)	29 h (18–52 h)
Hepatic metabolism	40 % reduction 60 % oxidation 1A2 > 3A4 > 2C19	10 % reduction 90 % oxidation 2C9 > 3A4
Stereospecific potency	1.0 (reference)	2.7–3.8 × R warfarin

Warfarin (Cont.)

- **Genetic polymorphisms** that involve response variability
 - **CYP2C9 poor metabolizers**
 - CYP2C9*2 (144R → C)
 - CYP2C9*3 (359I → L)
 - **VKORC1 (1639G → A)**
 - More sensitive to warfarin
- **Serious ADRs (1-10%)**
 - Purple toe syndrome
 - **Warfarin-induced skin necrosis**
 - A paradoxical reaction
 - Occur in the first few days of warfarin therapy (on the 3rd -8th day)



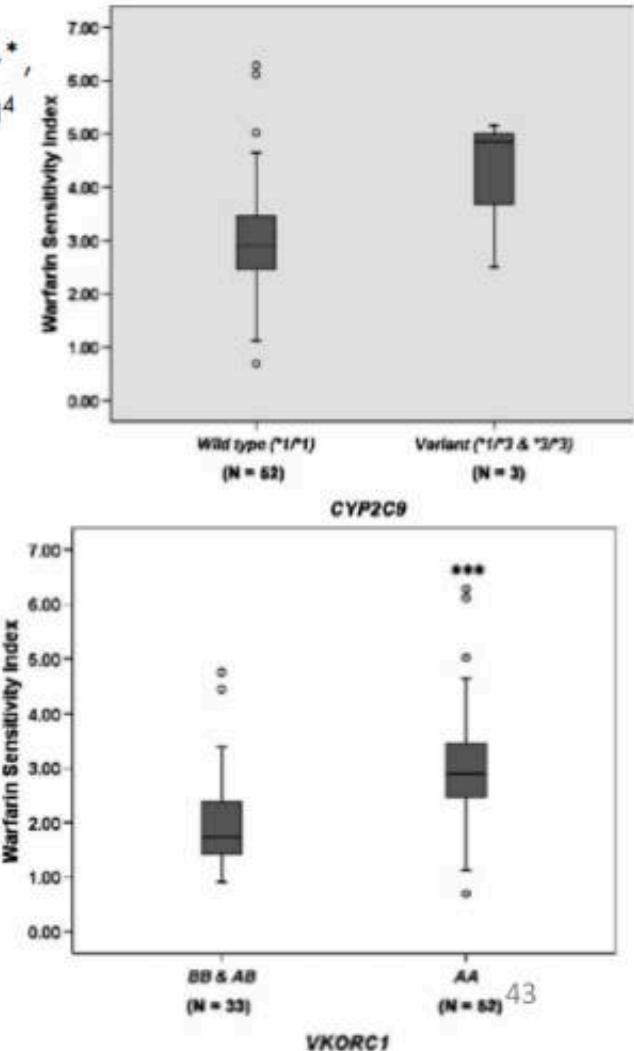
Regular Article

Pharmacokinetic and Pharmacodynamic Variation Associated with *VKORC1* and *CYP2C9* Polymorphisms in Thai Patients Taking Warfarin

Alisara SANGVIROON^{1,2}, Duangchit PANOMVANA^{1,*},
Wichitra TASSANEYAKUL³ and Jule NAMCHASIRI⁴

(89 patients × 2 alleles)				
	Alleles	N = 178	%	[95%CI]
<i>CYP2C9</i>	*1	173	97.2	[93.6–98.8]
	*2	0	0	—
	*3	5	2.8	[1.2–6.4]
<i>VKORC1</i> 1173C>T	C	40	22.5	[17.0–29.2]
	T	138	77.5	[70.9–83.0]
<i>VKORC1</i> – 1639G>A	G	40	22.5	[17.0–29.2]
	A	138	77.5	[70.9–83.0]

AA (1173TT, – 1639AA)	AB (1173CT, – 1639GA)	BB (1173CC, – 1639GG)
--------------------------	--------------------------	--------------------------





Drug Interactions with Warfarin

Can increase the INR

Acetaminophen³⁵⁻³⁷

Amiodarone^{38,39}

Cimetidine^{40,41}

Ciprofloxacin⁴²

Clarithromycin^{43,44}

Clofibrate^{45,46}

Erythromycin^{47,48}

Fluconazole^{49,50}

Fluvastatin^{51,52}

Fluvoxamine⁵³

Gemfibrozil⁵⁴

Isoniazid⁵⁵

Itraconazole⁵⁶

Ketoconazole⁵⁷

Lovastatin⁵⁸

Metronidazole⁵⁹

NSAIDs^{76,77}

Propafenone⁶⁰

Simvastatin⁶¹

Tamoxifen^{62,63}

Thyroid hormones⁷⁵

Tramadol^{64,65}

Trimethoprim

sulfamethoxazole⁶⁶

Zafirlukast⁶⁷

Unknown*

2C9 inhibition†

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

2C9 inhibition†

2C9 inhibition

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

2C9 inhibition†

Platelet inhibition‡

Enzyme inhibition

Enzyme inhibition

Enzyme inhibition

Increased vitamin K catabolism

Enzyme inhibition

2C9 inhibition†

2C9 inhibition

Can decrease the INR

Barbiturates^{68,69}

Carbamazepine⁷⁰

Dicloxacillin⁷¹

Methimazole⁷⁵

Propylthiouracil⁷⁵

Rifampin⁷⁴

Initially increase,
then decrease the INR

Phenytoin^{72,73}

Enzyme induction

Enzyme induction

Enzyme induction

Decreased vitamin K catabolism

Decreased vitamin K catabolism

Enzyme induction

Protein displacement/
enzyme induction

*Lesser effect

†Greater effect

‡Only reported with certain NSAIDs and appears to be dose-related



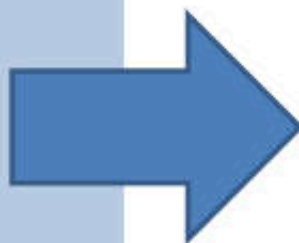
Direct-acting Anticoagulants

- **Thrombin inhibitors**

- Injectable drugs
 - Desirudin and lepirudin
 - Bivalirudin
 - Argatroban
- Oral drugs
 - Xymelagatran (withdrawn due to hepatotoxicity)
 - **Dabigatran etexilate**

- **FXa inhibitors**

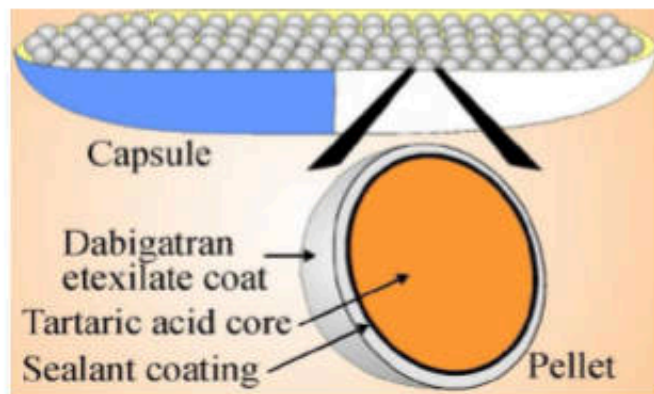
- Rivaroxaban
- Apixaban
- Edoxaban
- Betrixaban



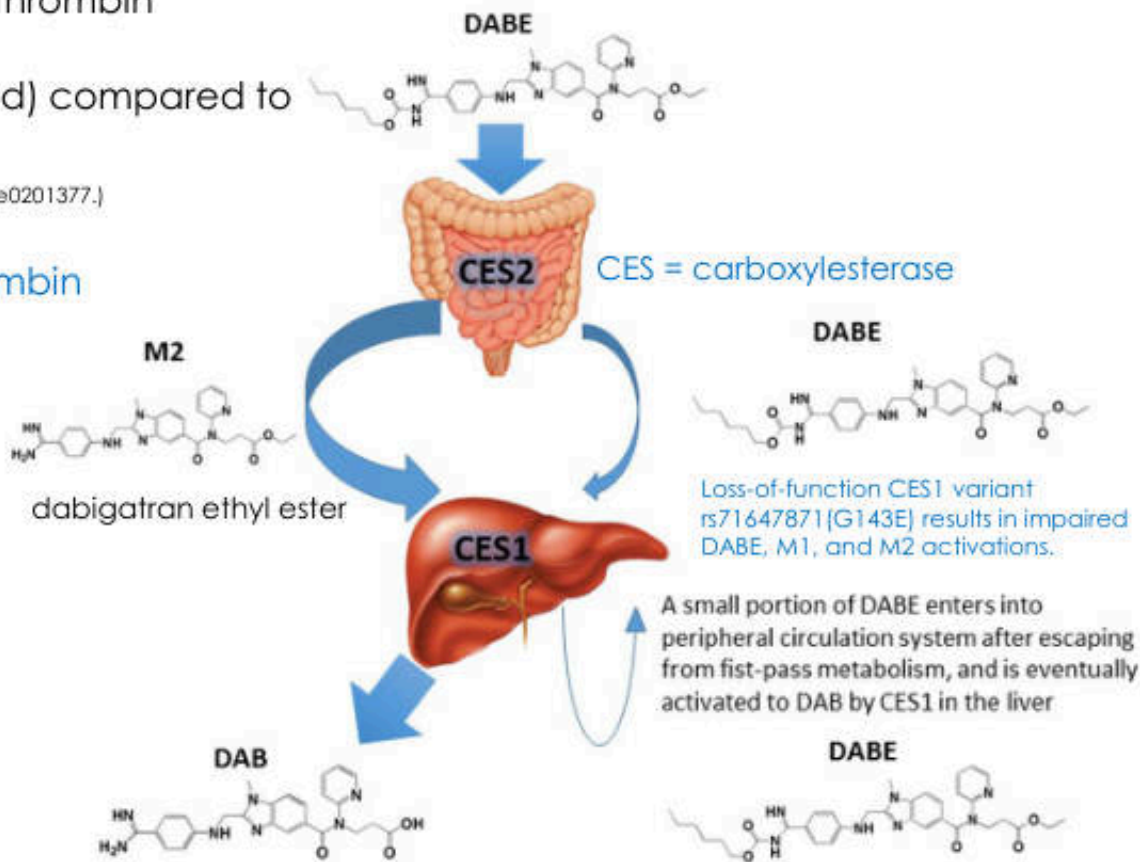
- **Direct-acting oral anticoagulants (DOACs)**
- or
- **Non-vitamin K antagonist oral anticoagulants (NOACs)**

Dabigatran etexilate

- Dabigatran is highly polar → dabigatran etexilate (prodrug)
- The only orally active direct thrombin inhibitor
- Competitive reversible inhibition of thrombin
- Highly specific to thrombin (>100-fold) compared to other serine proteases (PLoS ONE. 2018;13(8): e0201377.)
- Bind both free and clot-bound thrombin
- Dose-dependent action



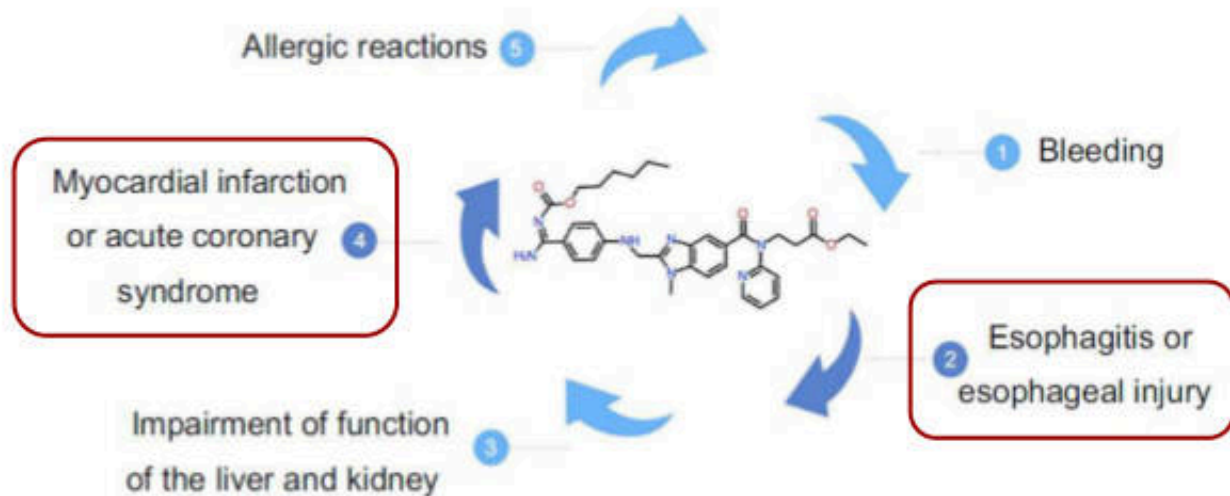
Can J Cardiol. 2018;34(9):1116-9.



Biochem Pharmacol. 2006;119:76-84.

ADRs of Dabigatran

Adverse events of dabigatran



Drug Design, Development and Therapy. 2019;13:1527-33.

- It has been speculated that the tartaric acid adheres to the esophagus to damage the esophageal mucosa
- If heartburn, chest pain and upper abdominal pain appear shortly after dabigatran administration,
- Drinking a sufficient volume of water (150 mL or more)
- Maintaining an upright position for ≥ 30 mins are useful strategies for relieving symptoms.
- If necessary, PPIs and H2RAs can be added.
- If symptoms persist, dabigatran should be discontinued
- Apixaban, rivaroxaban and warfarin are alternatives

The RE-LY trial showed that the prevalence of MI per year

- 0.53% with warfarin
- 0.72% with 110 mg of dabigatran
- 0.74% with 150 mg of dabigatran

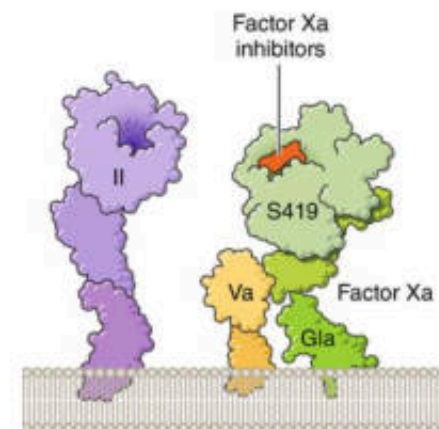
Direct factor Xa inhibitors

- All of them are competitive reversible inhibitors
- Bind both free and clot-bound FXa
- Dose-dependent action
- Highly specific to FXa compared with other serine proteases

	FXa specificity (fold; IC ₅₀ ratio)
Rivaroxaban ¹	≈10,000
Apixaban ²	> 10,000
Edoxaban ³	> 10,000
Betrixaban ⁴	> 86,000

¹Clin Pharmacokinet. 2014;53:1-16. ²J Thromb Thrombolysis. 2011;31:478-92.

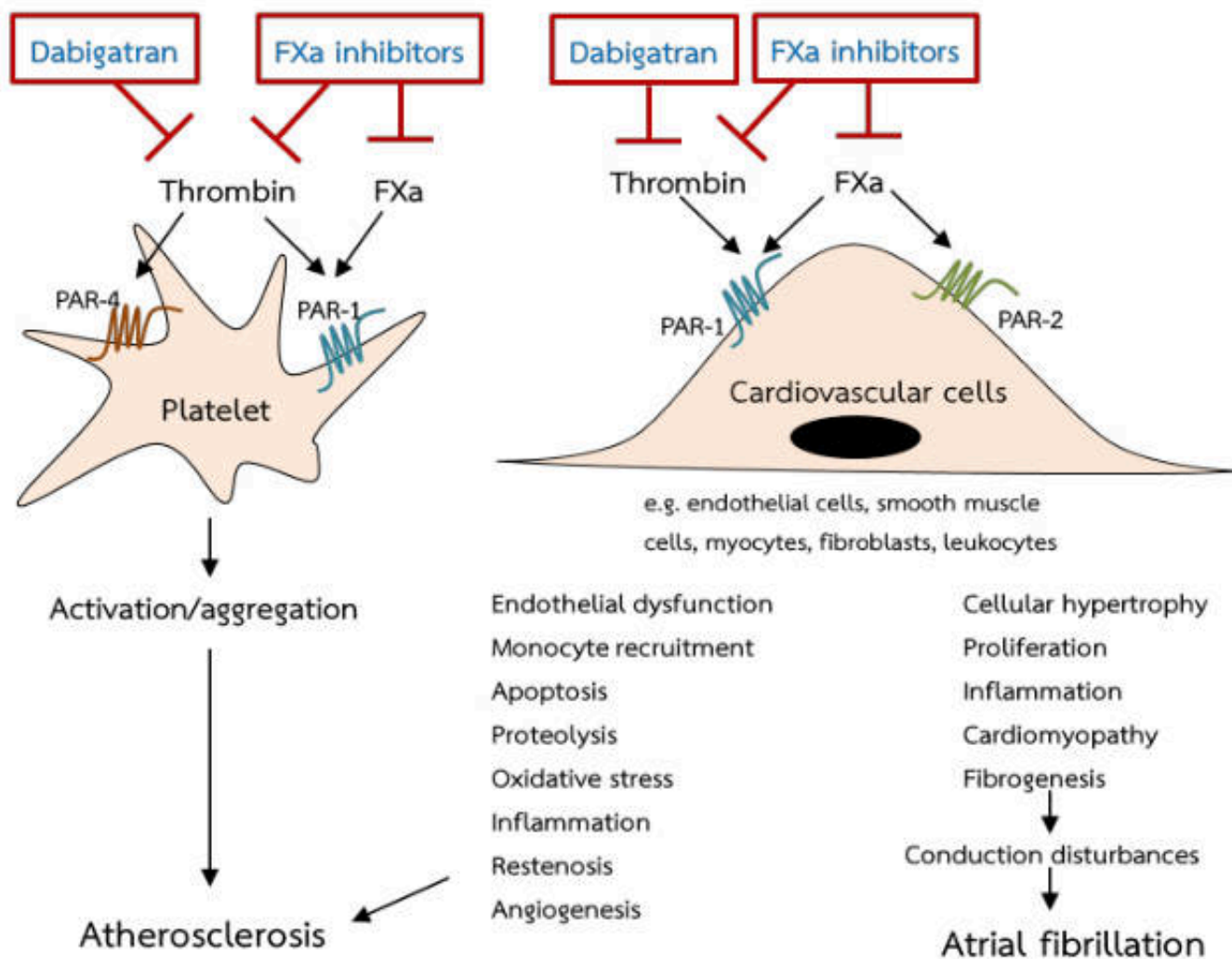
³Clin Pharmacokinet. 2016;55:641-55. ⁴Bioorg Med Chem Lett. 2009;19:2179-85.



Circulation. 2016;134:248–61.



Antiplatelet effect and other potential benefits of NOACs





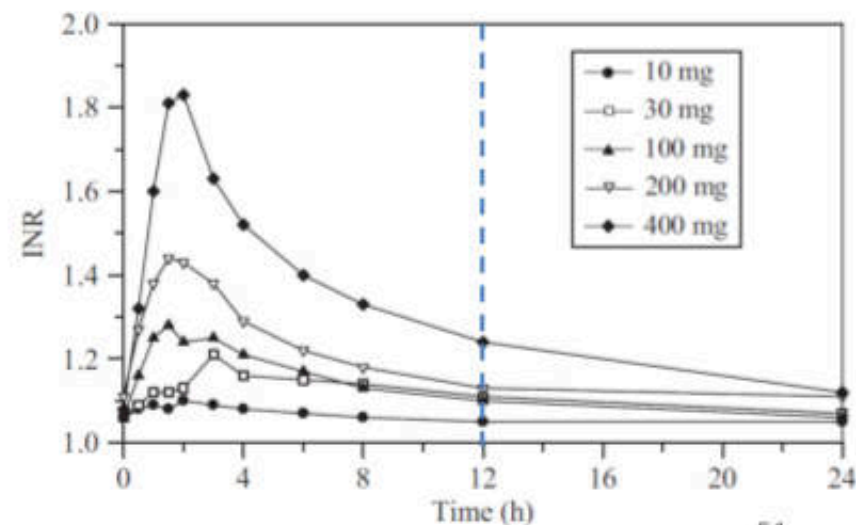
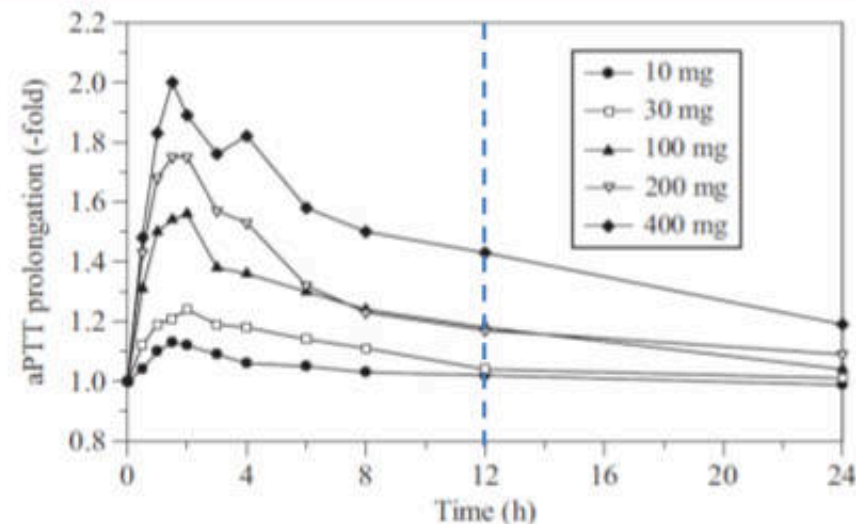
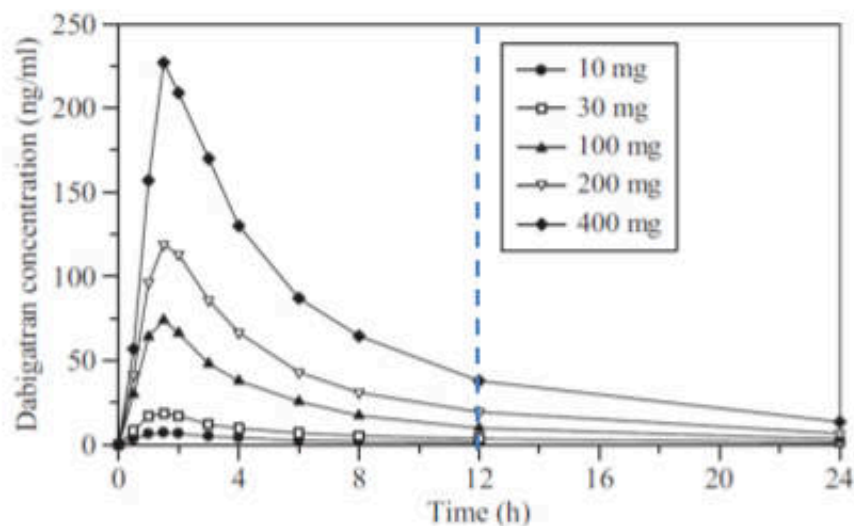
Pharmacokinetic parameters of NOACs

	Dabigatran	Rivaroxaban	Apixaban	Edoxaban	Betrixaban
Bioavailability (%)	6-7	80-100	50	62	34
Tmax (h)	1.5-3	2-4	1-3	1-2	3-4
Tmin (time to C _{trough} , h)	12	24 12 (at ≤ 5 mg)	12	24	24
Duration of action (h)	12 ^a	24 ^b 12 (at ≤ 5 mg)	12 ^b	24 ^b	24 ^b
Half-life (h)	12-17	7-11 5-7 (at ≤ 5 mg)	8-14	10-14	19-27
Protein binding (%)	30-35	90-95	87	40-59	60
Volume of distribution	50-70 L	50 L	23 L	>300 L	32 L/kg
Metabolism via CYP	< 2%	CYP3A4 (major, CYP2J2	CYP3A4	< 4%	< 1%
Renal excretion as unchanged (%)	80	33	27	35-50	11
	----- Dose reduction in renal impairment -----				
P-gp substrate	Yes	Yes	Yes	Yes	Yes

^aduration for inhibition of thrombin activity, ^bduration for inhibition of thrombin generation



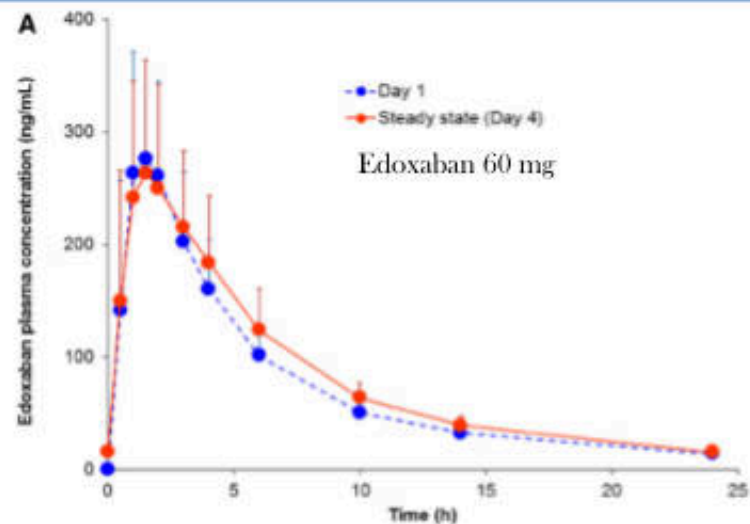
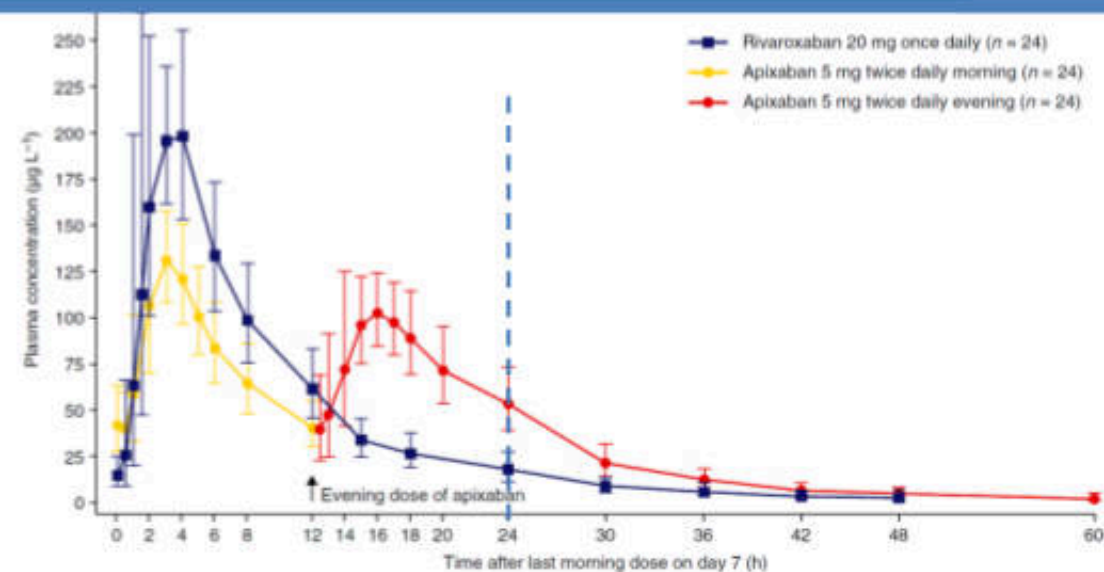
Clinical pharmacokinetics and pharmacodynamics of dabigatran



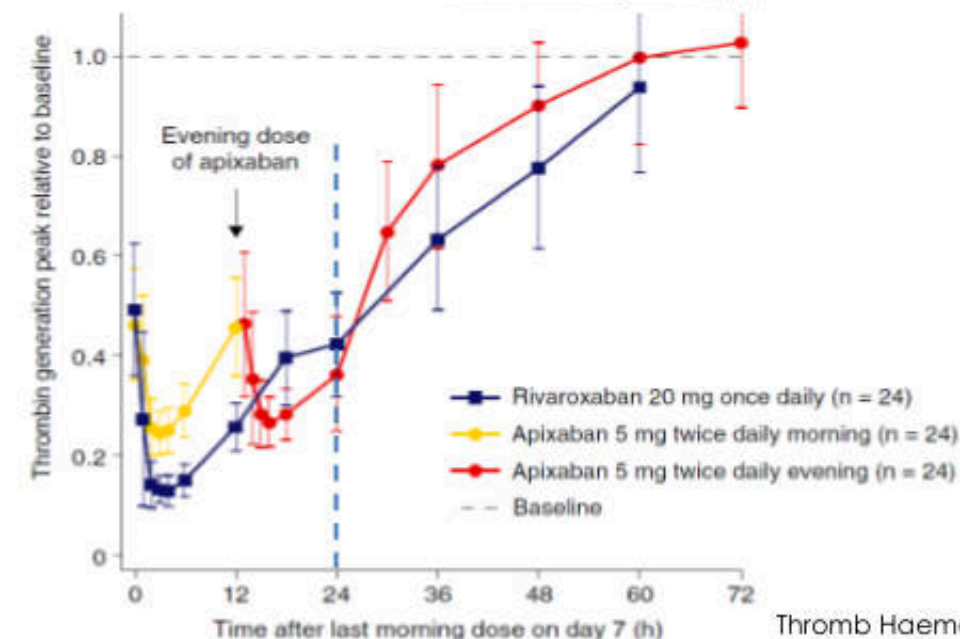
Usual dosing: 110-150 mg BID



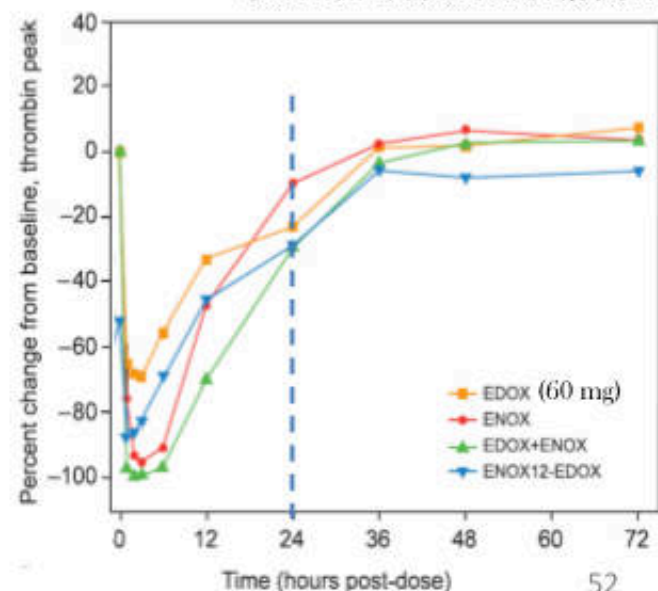
Clinical pharmacokinetics and pharmacodynamics of FXa inhibitors



Clin Pharmacokinet. 2016;55:641–55.



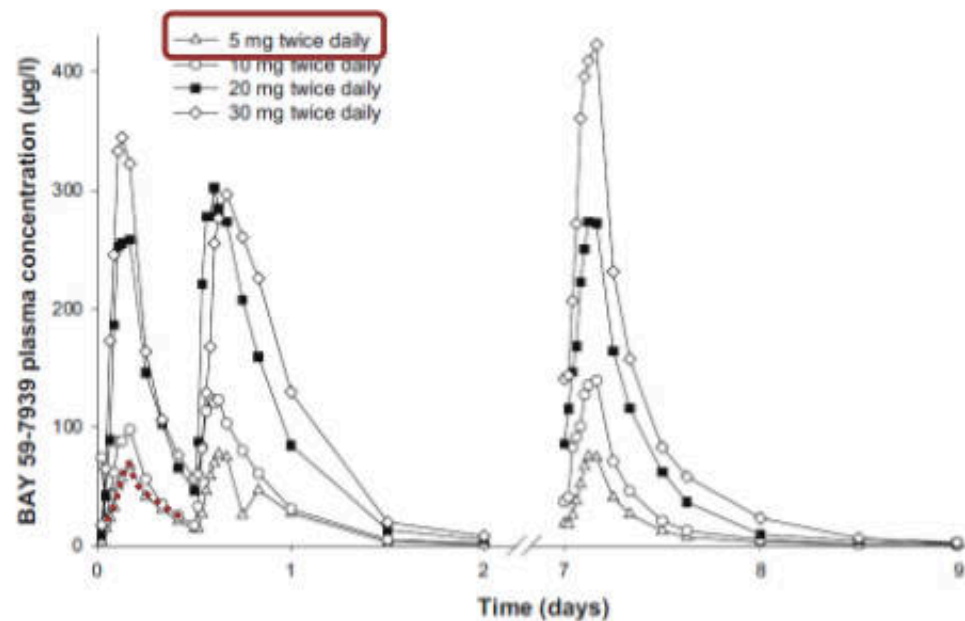
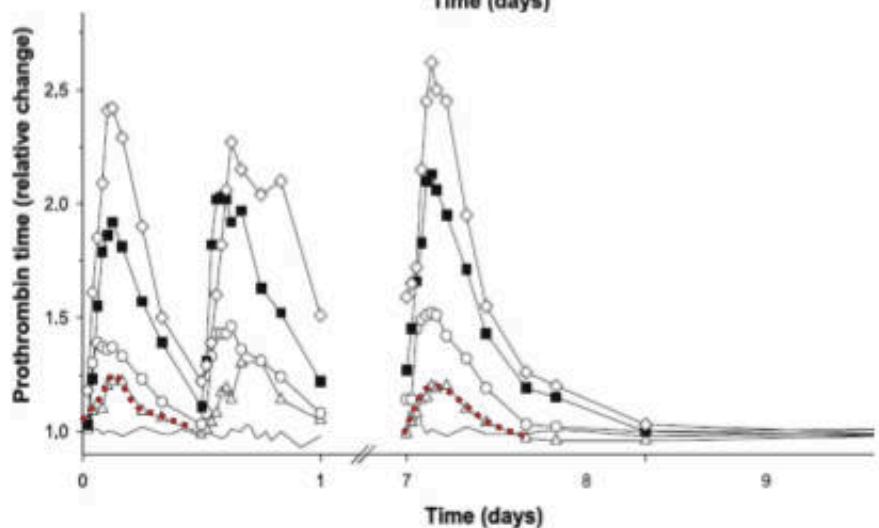
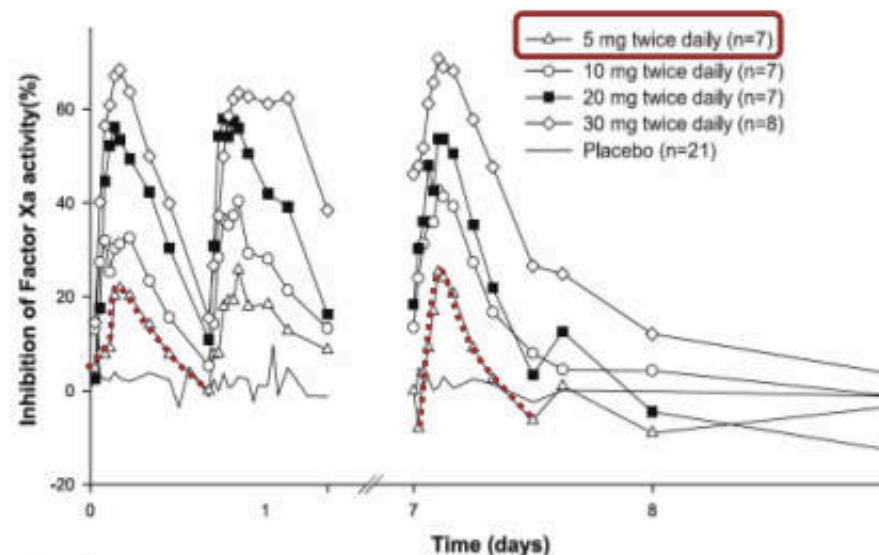
Thromb Haemost. 2017; 15: 2017–28.



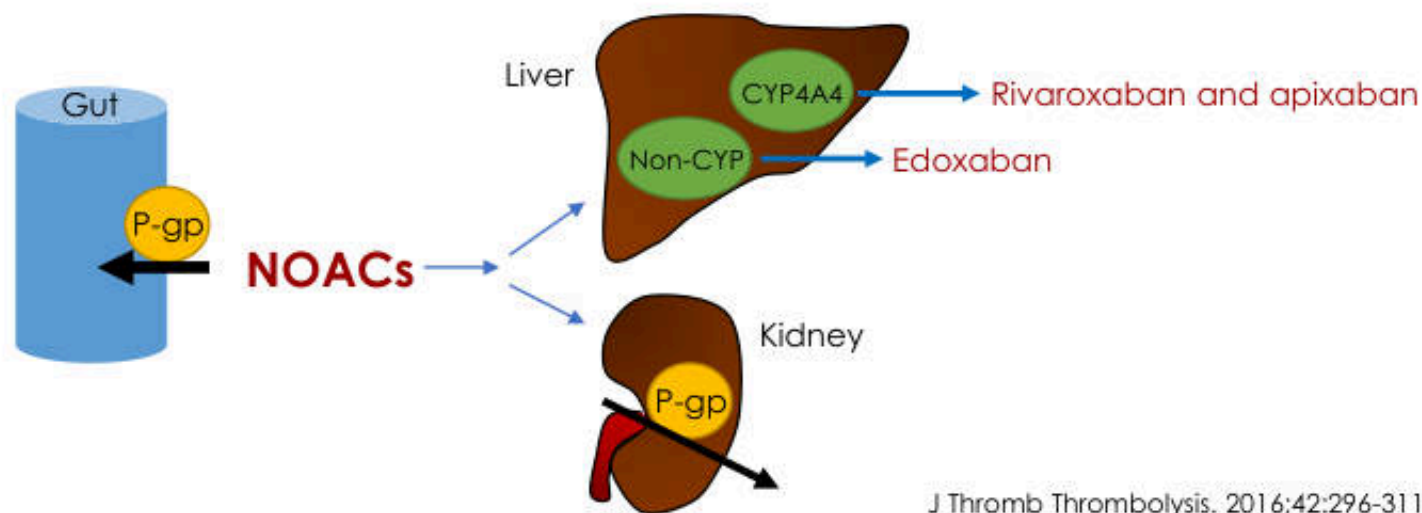
Thromb Haemost. 2012; 108: 166–75.



Clinical pharmacokinetics and pharmacodynamics of low-dose of rivaroxaban (≤ 5 mg)



Pharmacokinetic drug interactions of NOACs



P-gp inducers	P-gp and CYP3A4 inducers	P-gp inhibitors	P-gp and CYP3A4 inhibitors	
Dexamethasone	Barbiturates, Carbamazepine	Amiodarone, Quinidine	Verapamil , Diltiazem	Ketoconazole
	Phenytoin	Propafenone	Dronedarone	Itraconazole
	Rifampin	Carvedilol, Nicardipine	Clarithromycin	Posaconazole
	St. John's Wort	Lepatinib	Erythromycin	Nelfinavir, Ritonavir
		Tacrolimus	Cyclosporine	Saquinavir, Indinavir
		Mefloquine	Tamoxifen, Conivaptan	Grapefruit



Decrease systemic exposure of NOACs

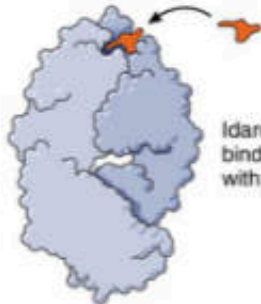
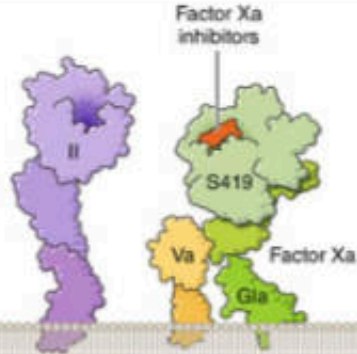
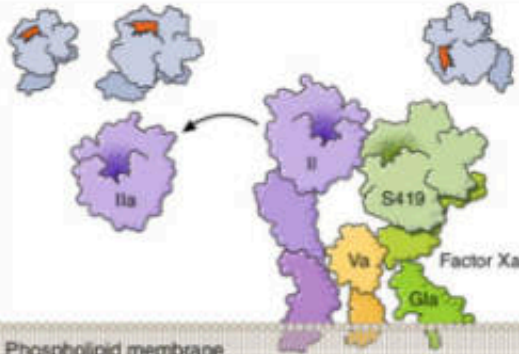
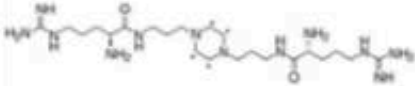
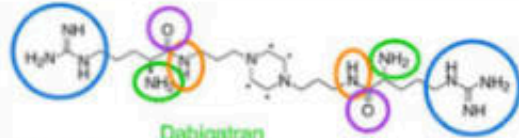


Increase systemic exposure of NOACs

Specific antidotes for NOACs

Currently approved for

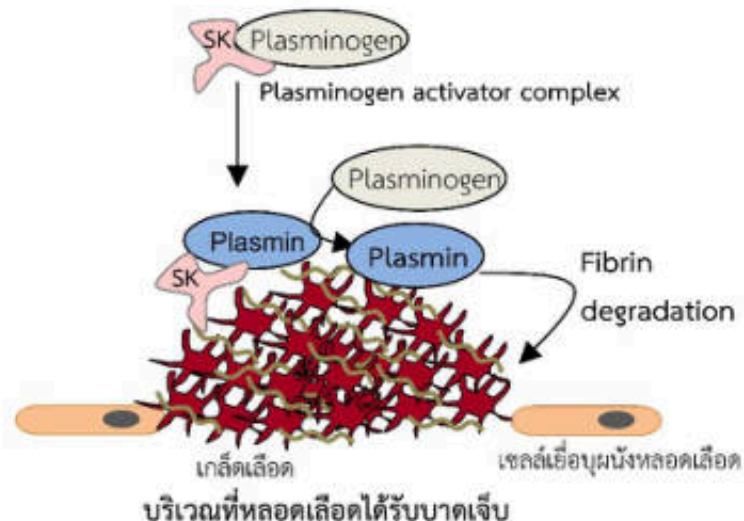
- Rivaroxaban
- Apixaban

NOAC reversal agent	Target	Mechanism
 Idarucizumab	 Dabigatran	 Idarucizumab binds Dabigatran with high affinity
 A419 Andexanet alpha	 Factor Xa inhibitors II S419 Va Gla Factor Xa Phospholipid membrane	 Phospholipid membrane
 Ciraparantag (PER977)	Apixaban Argatroban Edoxaban Dabigatran Rivaroxaban UFH LMWH Fondaparinux	 Edoxaban Dabigatran Rivaroxaban Apixaban Argatroban UFH/LMWH Fondaparinux Computer-aided energy minimization modeling predicts 8 non-covalent binding sites on ciraparantag for NOACs or heparins

Thrombolytics

- Streptokinase

- The enzyme derived from β -hemolytic *Streptococci*



- Alteplase

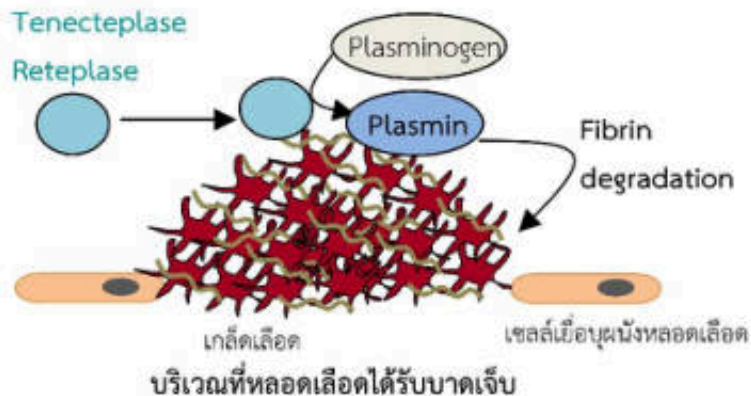
- Recombinant human tPA

Urokinase

Alteplase

Tenecteplase

Reteplase



Wichaiyo S.



Comparative Pharmacological Properties of Thrombolytics

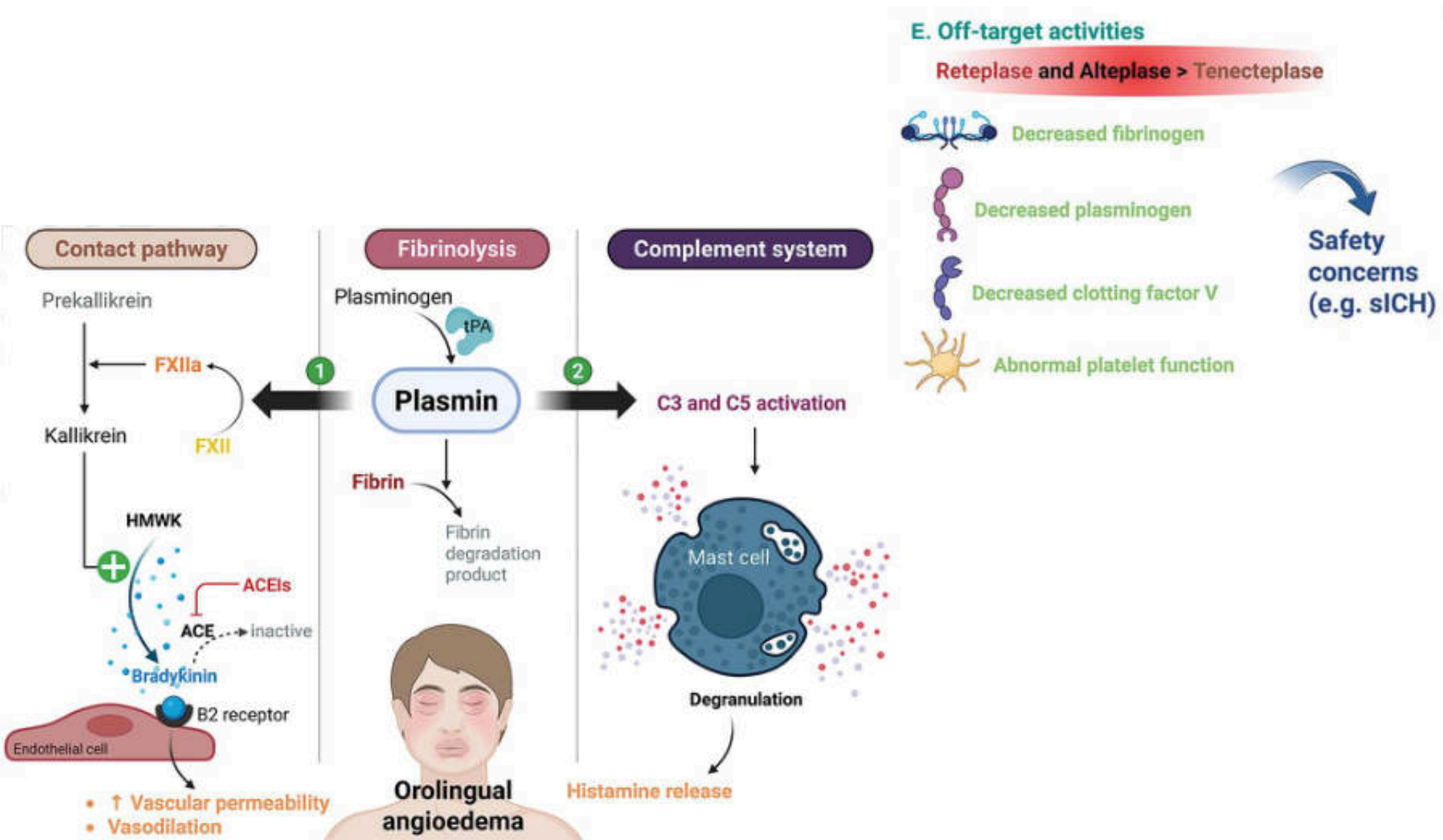
	Streptokinase	Alteplase	Tenecteplase	Reteplase
Action	Indirect	Direct	Direct	Direct
Fibrin Specificity	Non-specific	++	+++	+
Clot penetration		+	+	+++
PAI-1 resistance		++	++++	+
t1/2 (min)	20	<5	20	13-16
Excretion	Liver	Liver	Liver	Liver & Kidney
Hypersensitivity	Yes (Elevated level of antibody against streptokinase persists for up to 4 years)	No	No	No
Indication(s)	ACS, PE 	• Acute ischemic stroke • ACS • Acute PE • Central venous catheter occlusion	• Acute ischemic stroke • ACS	ACS

¹Semin Intervent Radiol. 2010;27(4):374-383. ²Drugs Aging. 1994;4(1):63-86. ³J Stroke. 2013;15(2):90-98.

⁴Biochem Pharmacol. 2013;85(2):216-222. ⁵N Engl J Med 2024;390:2264-2273.



Serious ADRs of thrombolytics





Clinical Uses of Antithrombotics

- **Prophylaxis/primary prevention**
- **Treatment**
- **Secondary prevention/prevention of recurrence**

Further reading

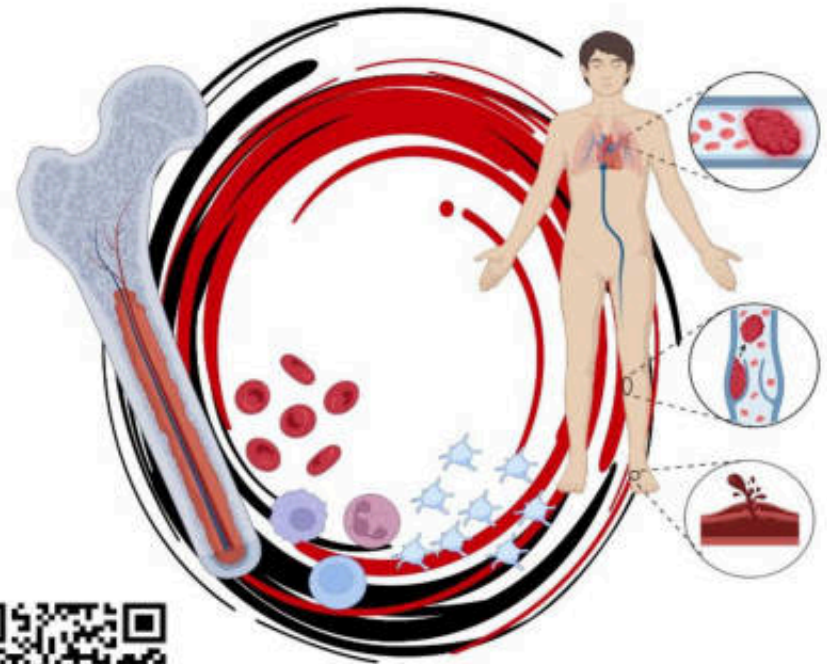


เภสัชวิทยาของยารักษาโรคทางโลหิตวิทยา
ที่ไม่ใช่มะเร็ง

Pharmacology of Drugs Used in Nonmalignant
Hematological Disorders

Thank You

Any Questions?



สุรศักดิ์ รัชชโย
บรรณาธิการ